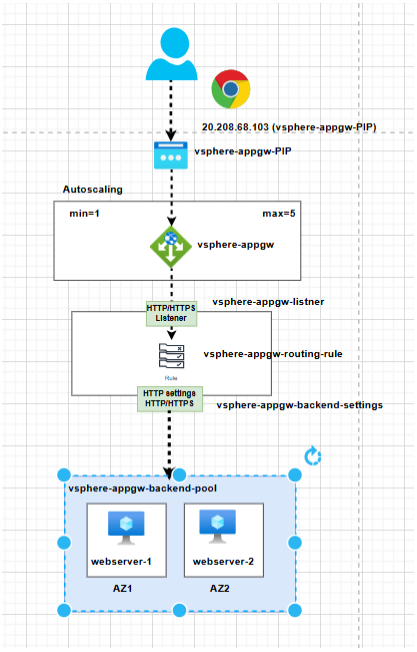
Lab 15 – Azure Application Gateway with Web Servers, Certificate, and DNS Zone

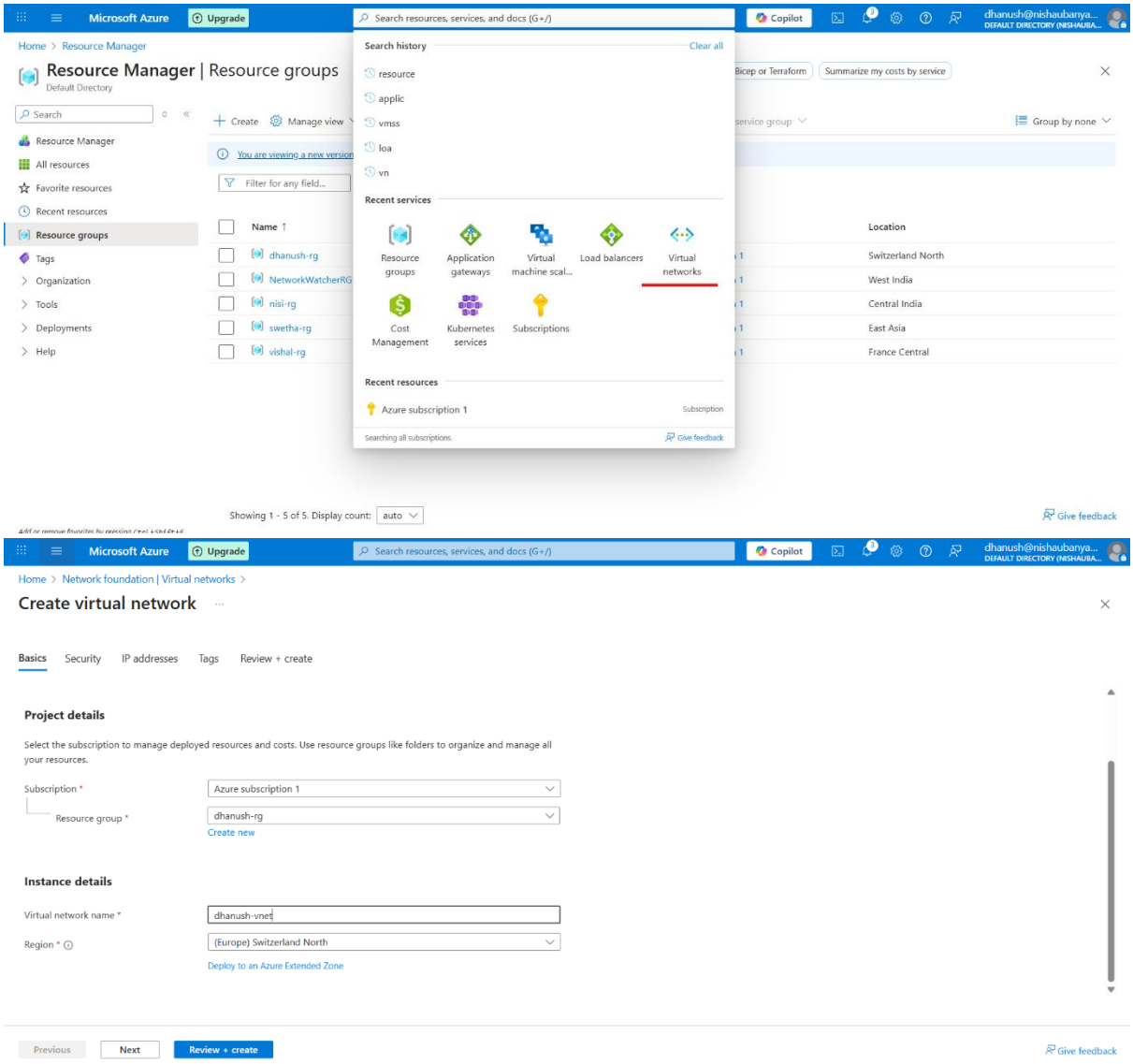
Goal

Deploy two Linux web servers, configure an Application Gateway with HTTP & HTTPS listeners, create a PFX certificate, add a DNS zone for friendly access, and test both HTTP & HTTPS traffic.



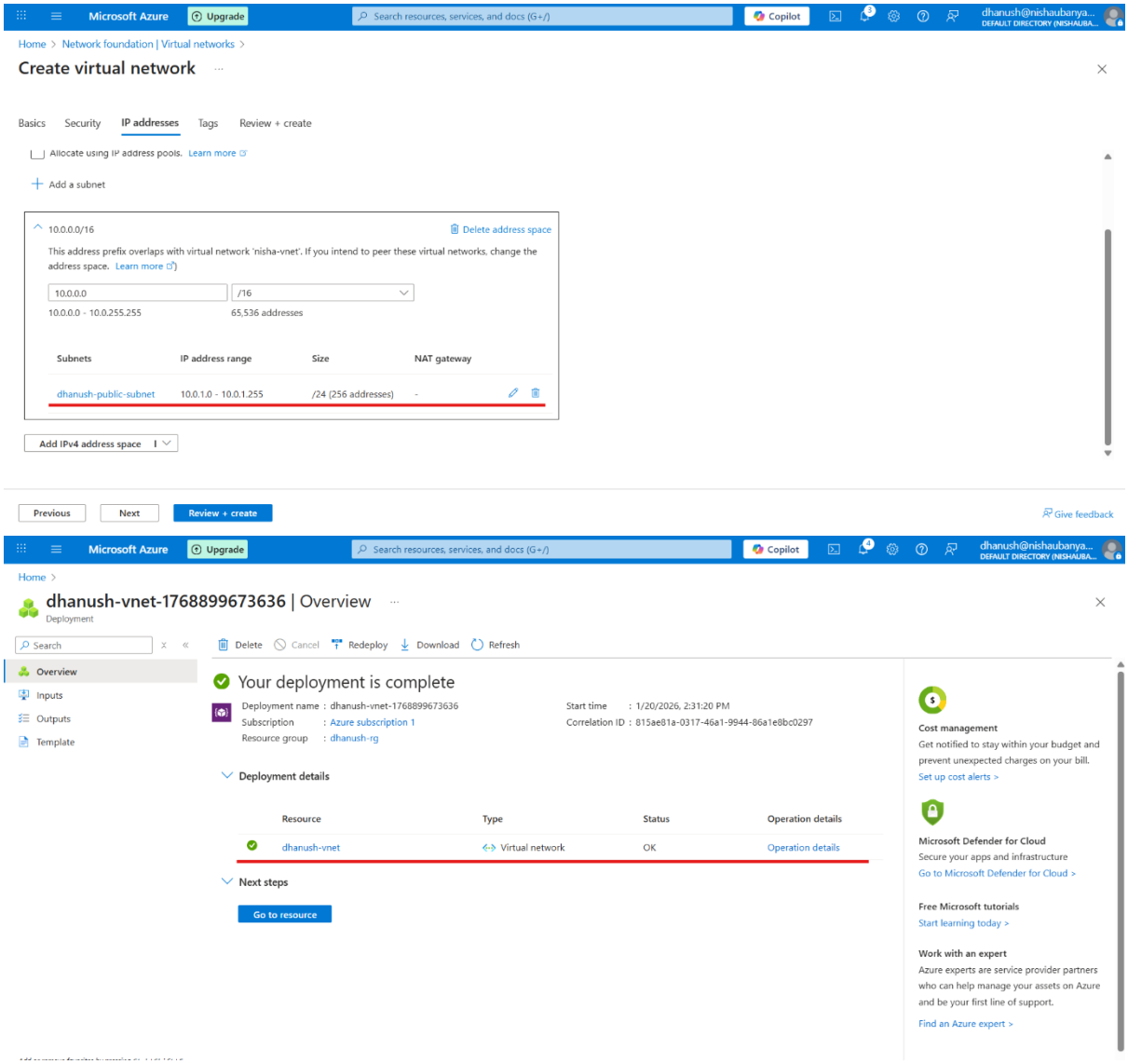
STEP 1 : Create Resource Group

STEP 2: Create Virtual Network


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Copilot
dhanush@nishaubanya...
DEFAULT DIRECTORY (NISHAUBA...
Home > Resource Manager
Resource Manager | Resource groups
Default Directory
Search
Create
Manage view
Group by none
Resource Manager
All resources
Favorite resources
Recent resources
Resource groups
Tags
Organization
Tools
Deployments
Help
Name
Location
dhanush-rg
Switzerland North
NetworkWatcherRG
West India
nisi-rg
Central India
swetha-rg
East Asia
vishal-rg
France Central
Search history
Clear all
resource
applic
vmss
loa
vn
Recent services
Resource groups
Application gateways
Virtual machine scal...
Load balancers
Virtual networks
Cost Management
Kubernetes services
Subscriptions
Recent resources
Azure subscription 1
Subscription
Searching all subscriptions.
Give feedback
Summarize my costs by service
Showing 1 - 5 of 5. Display count: auto
Give feedback

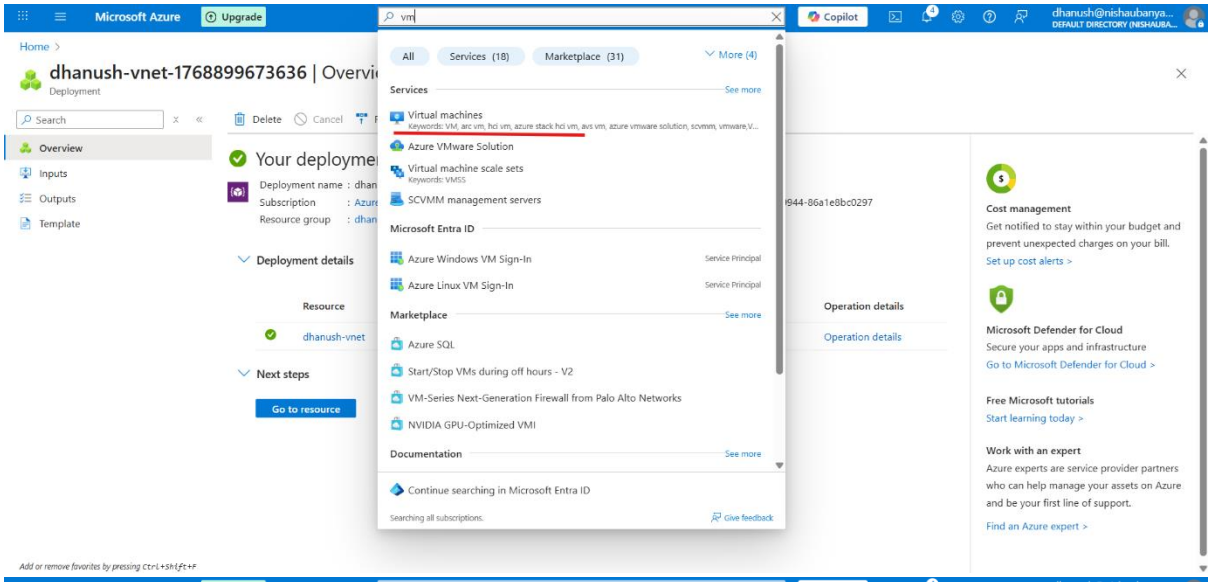
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DEFAULT DIRECTORY (NISHAUBA...
Home > Network foundation | Virtual networks >
Create virtual network
Basics
Security
IP addresses
Tags
Review + create
Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Azure subscription 1
Resource group *
dhanush-rg
Create new
Instance details
Virtual network name *
dhanush-vnet
Region *
(Europe) Switzerland North
Deploy to an Azure Extended Zone
Previous
Next
Review + create
Give feedback

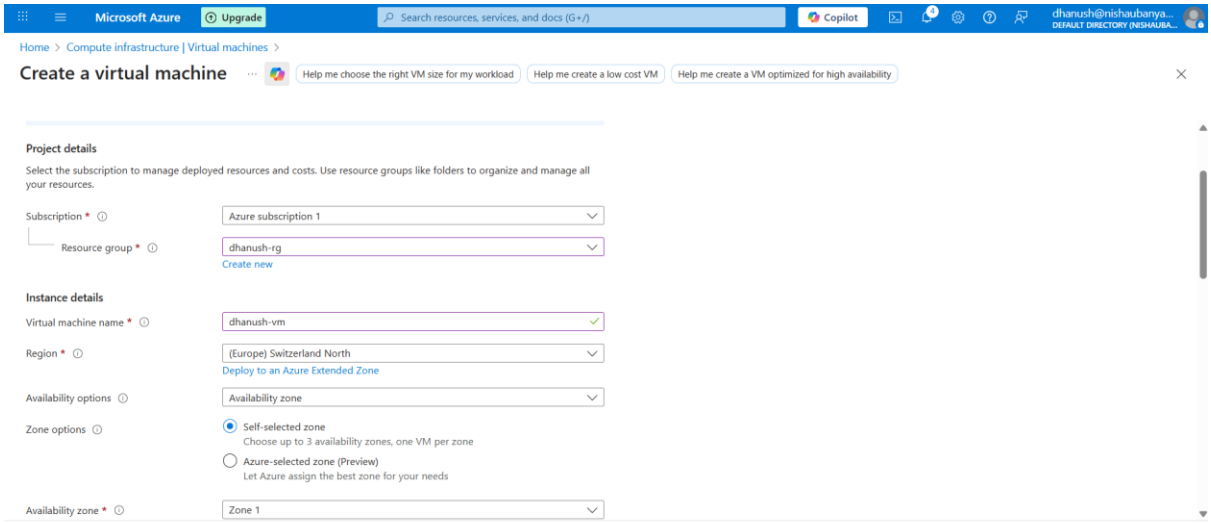
Add subnet

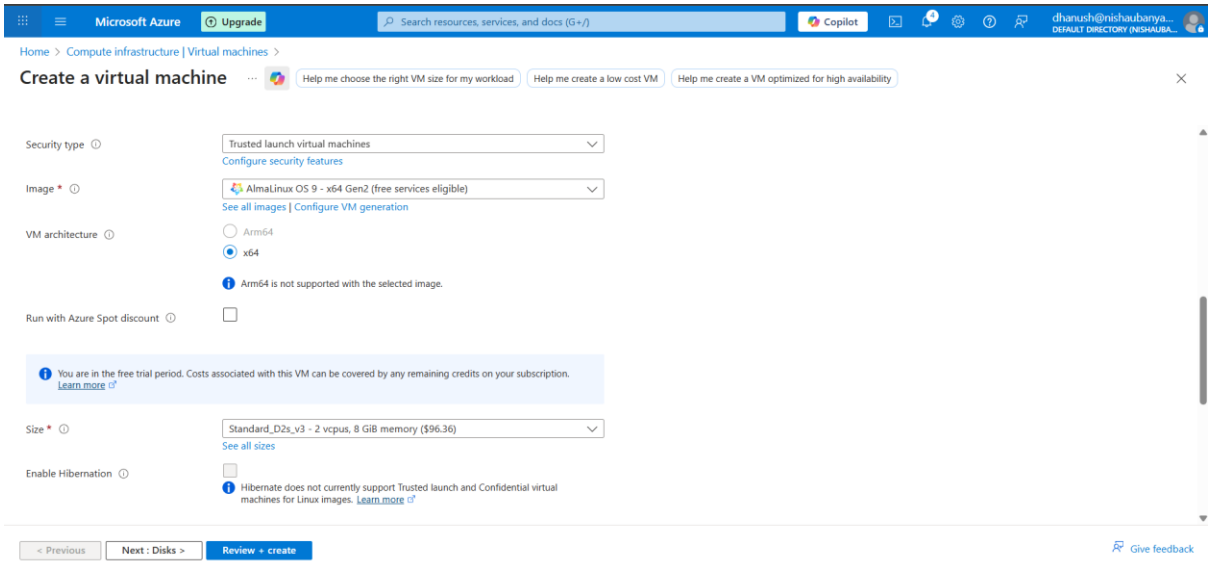


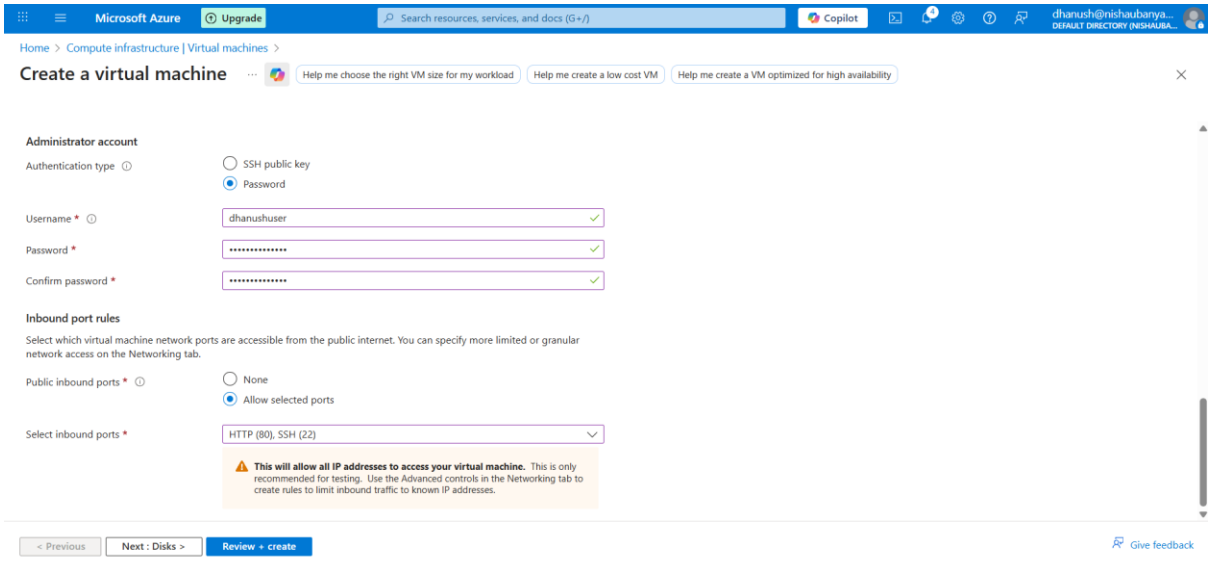
STEP 3 : Create Two Linux VMs

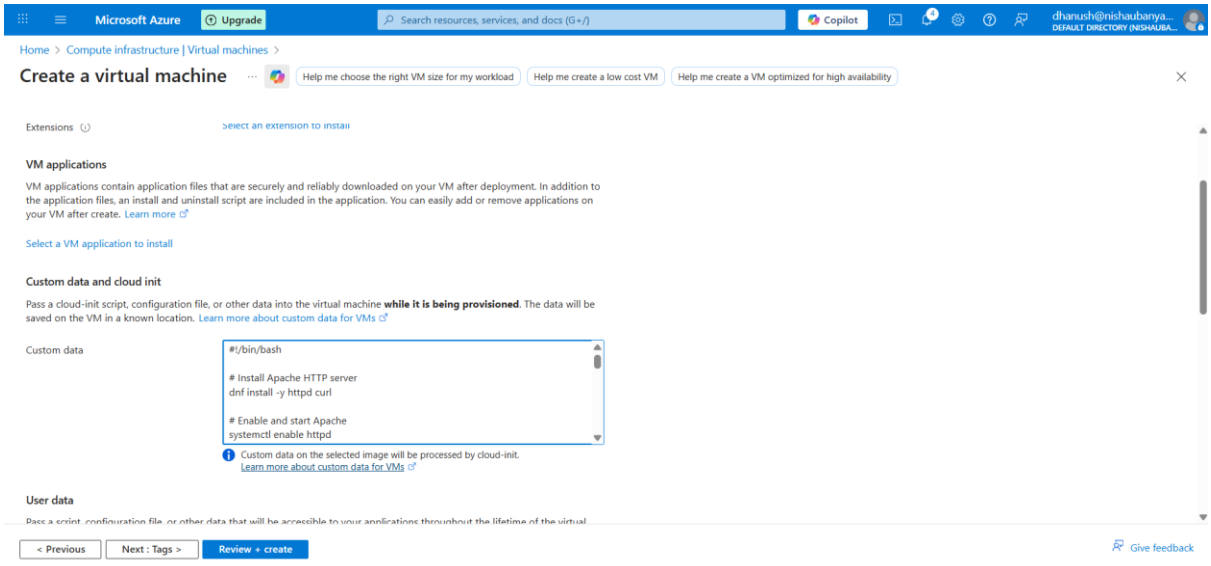
VM 1

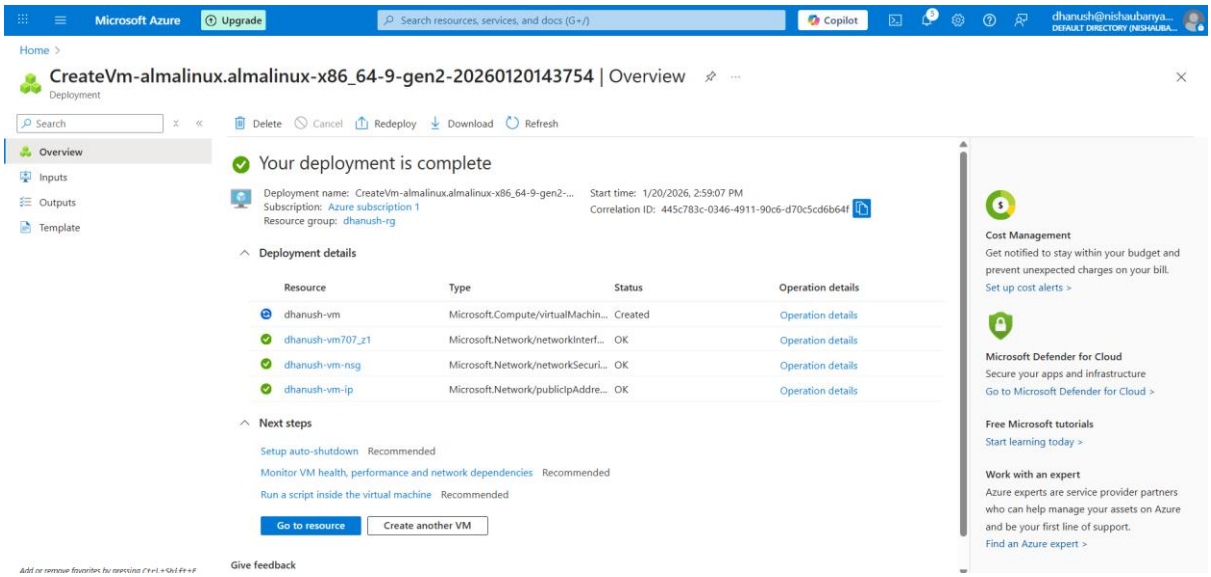

Microsoft Azure
Upgrade
vm
Copilot
dhanush@nishaubanya...
DEFAULT DIRECTORY (NISHAUBA...
Home >
dhanush-vnet-1768899673636 | Overvi
Deployment
Search
Delete
Cancel
Overview
Inputs
Outputs
Template
Your deployme
Deployment name : dhan
Subscription : Azur
Resource group : dhan
Deployment details
Resource
dhanush-vnet
Next steps
Go to resource
Operation details
Operation details
All
Services (18)
Marketplace (31)
More (4)
Services
See more
Virtual machines
Keywords: VM, arc vm, hci vm, azure stack hci vm, avs vm, azure vmware solution, scvmm, vmware,V...
Azure VMware Solution
Virtual machine scale sets
Keywords: VMSS
SCVMM management servers
Microsoft Entra ID
Azure Windows VM Sign-In
Service Principal
Azure Linux VM Sign-In
Service Principal
Marketplace
See more
Azure SQL
Start/Stop VMs during off hours - V2
VM-Series Next-Generation Firewall from Palo Alto Networks
NVIDIA GPU-Optimized VMI
Documentation
See more
Continue searching in Microsoft Entra ID
Searching all subscriptions.
Give feedback
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Set up cost alerts >
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Home > Compute infrastructure | Virtual machines >
Create a virtual machine
Help me choose the right VM size for my workload
Help me create a low cost VM
Help me create a VM optimized for high availability
Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Azure subscription 1
Resource group *
dhanush-rg
Create new
Instance details
Virtual machine name *
dhanush-vm
Region *
(Europe) Switzerland North
Deploy to an Azure Extended Zone
Availability options
Availability zone
Zone options
Self-selected zone
Choose up to 3 availability zones, one VM per zone
Azure-selected zone (Preview)
Let Azure assign the best zone for your needs
Availability zone *
Zone 1
< Previous
Next : Disks >
Review + create
Give feedback

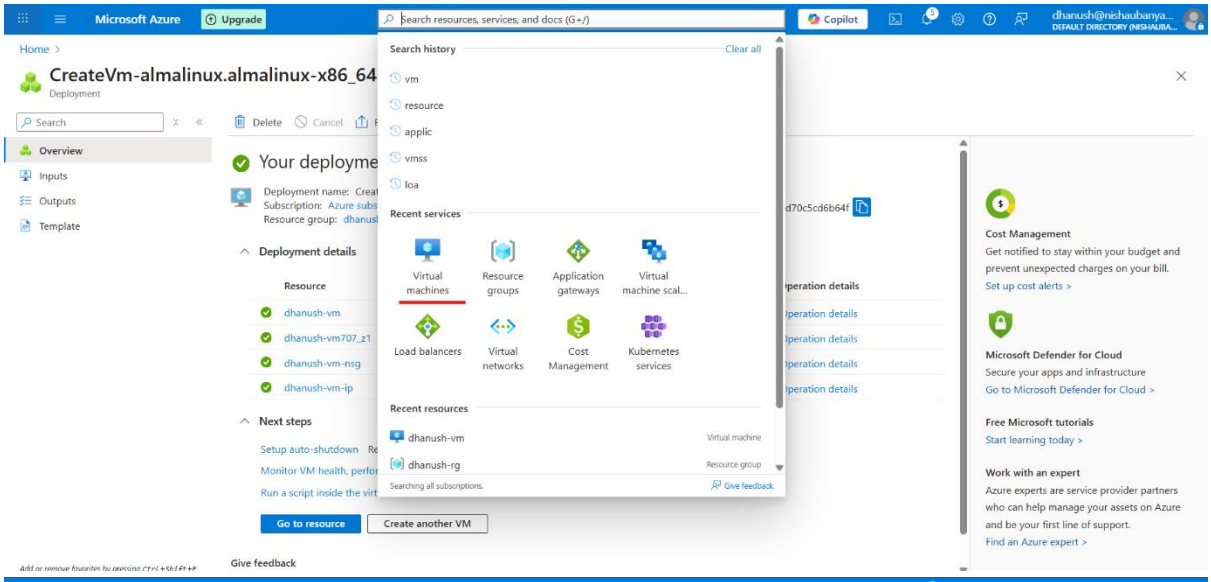

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Create a virtual machine
Help me choose the right VM size for my workload
Help me create a low cost VM
Help me create a VM optimized for high availability
Security type
Trusted launch virtual machines
Configure security features
Image *
AlmaLinux OS 9 - x64 Gen2 (free services eligible)
See all images | Configure VM generation
VM architecture
Arm64
x64
Arm64 is not supported with the selected image.
Run with Azure Spot discount
You are in the free trial period. Costs associated with this VM can be covered by any remaining credits on your subscription.
Learn more
Size *
Standard_D2s_v3 - 2 vcpus, 8 GiB memory ($96.36)
See all sizes
Enable Hibernation
Hibernate does not currently support Trusted launch and Confidential virtual machines for Linux images. Learn more
< Previous
Next : Disks >
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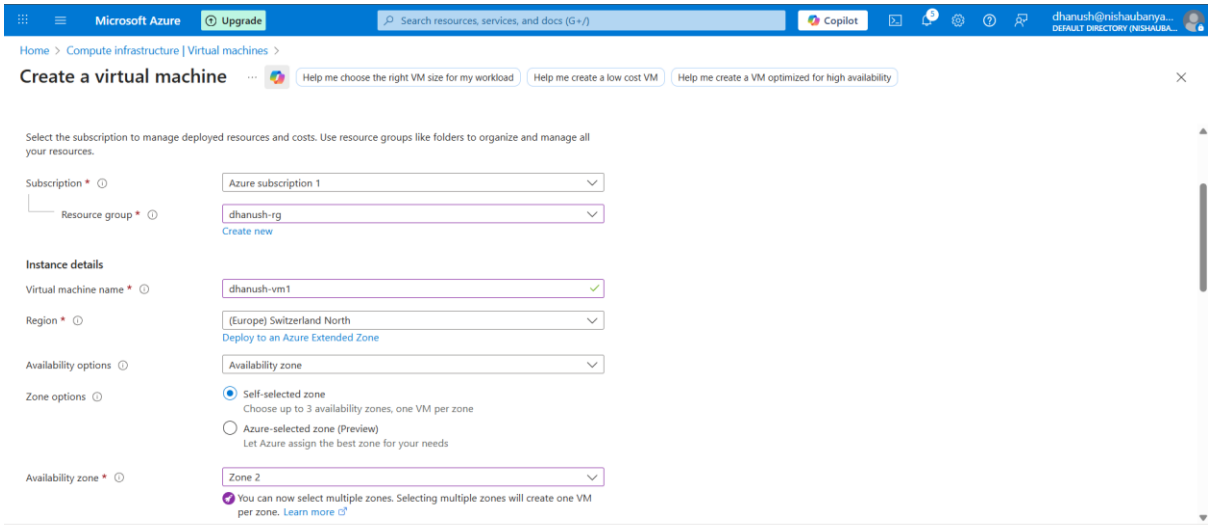

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DEFAULT DIRECTORY (NISHAUBA...
Home > Compute infrastructure | Virtual machines >
Create a virtual machine
Help me choose the right VM size for my workload
Help me create a low cost VM
Help me create a VM optimized for high availability
Administrator account
Authentication type
SSH public key
Password
Username *
dhanushuser
Password *
Confirm password *
Inbound port rules
Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.
Public inbound ports *
None
Allow selected ports
Select inbound ports *
HTTP (80), SSH (22)
This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.
< Previous
Next : Disks >
Review + create
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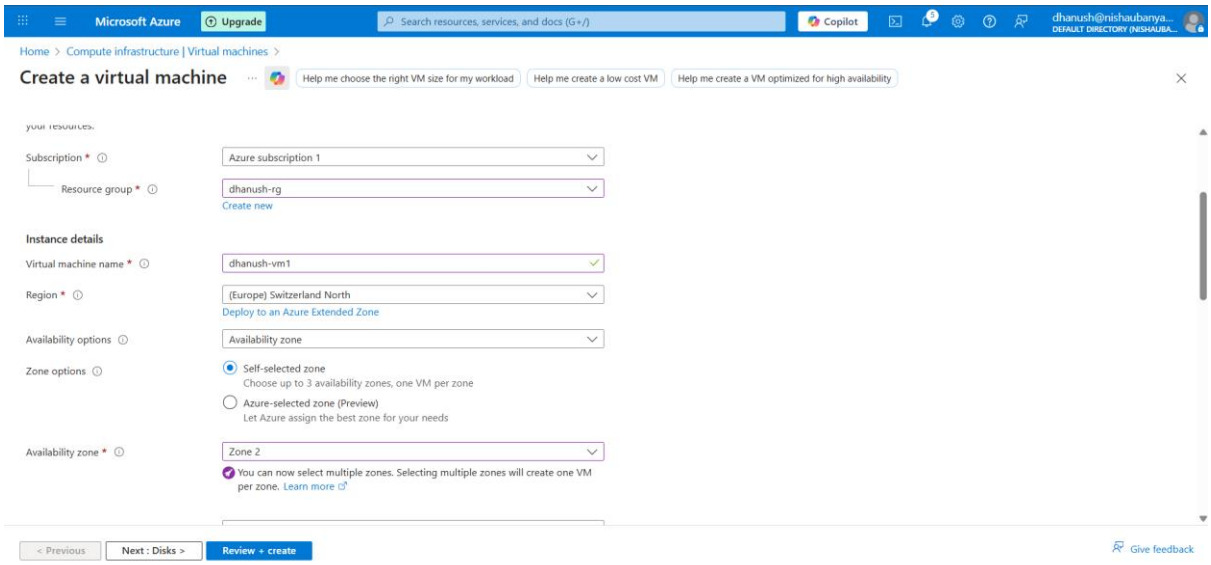

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DEFAULT DIRECTORY (NISHAUBA...
Home > Compute infrastructure | Virtual machines >
Create a virtual machine
Help me choose the right VM size for my workload
Help me create a low cost VM
Help me create a VM optimized for high availability
Extensions
VM applications
VM applications contain application files that are securely and reliably downloaded on your VM after deployment. In addition to the application files, an install and uninstall script are included in the application. You can easily add or remove applications on your VM after create. Learn more
Select a VM application to install
Custom data and cloud init
Pass a cloud-init script, configuration file, or other data into the virtual machine while it is being provisioned. The data will be saved on the VM in a known location. Learn more about custom data for VMs
Custom data
#!/bin/bash
Install Apache HTTP server
dnf install -y httpd curl
Enable and start Apache
systemctl enable httpd
Custom data on the selected image will be processed by cloud-init.
Learn more about custom data for VMs
User data
< Previous
Next : Tags >
Review + create
Give feedback

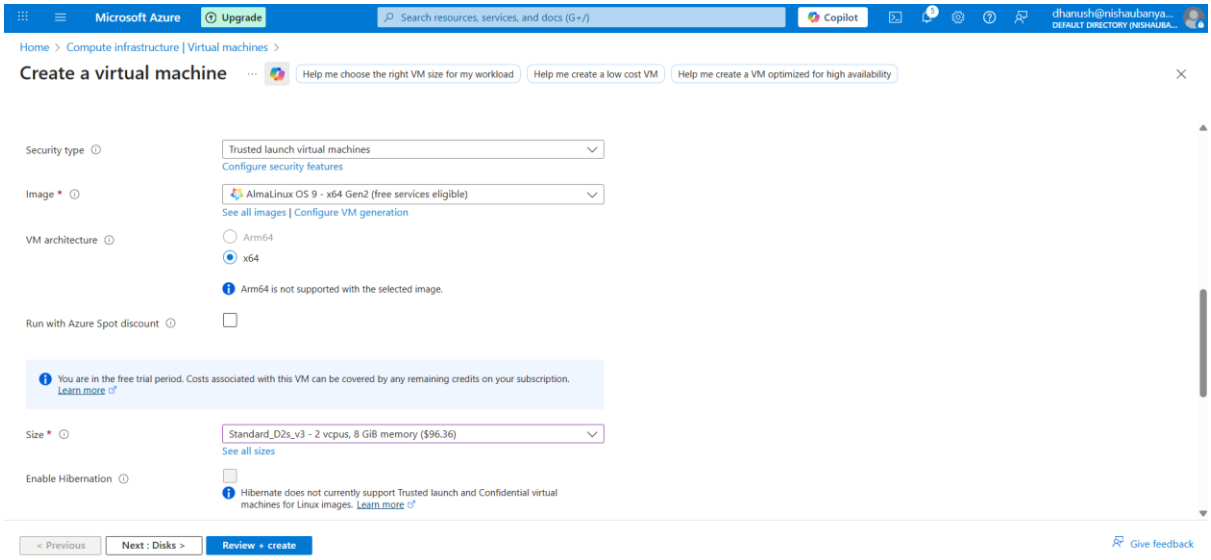

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Copilot
dhanush@nishaubanya...
DEFAULT DIRECTORY (NISHAUBA...
Home >
CreateVm-almalinux.almalinux-x86_64-9-gen2-20260120143754 | Overview
Deployment
Search
Overview
Inputs
Outputs
Template
Delete
Cancel
Redeploy
Download
Refresh
Your deployment is complete
Deployment name: CreateVm-almalinux.almalinux-x86_64-9-gen2-...
Subscription: Azure subscription 1
Resource group: dhanush-rg
Start time: 1/20/2026, 2:59:07 PM
Correlation ID: 445c783c-0346-4911-90c6-d70c5cd6b64f
Deployment details
Resource
Type
Status
Operation details
dhanush-vm
Microsoft.Compute/virtualMachin...
Created
Operation details
dhanush-vm707_z1
Microsoft.Network/networkInterf...
OK
Operation details
dhanush-vm-nsg
Microsoft.Network/networkSecuri...
OK
Operation details
dhanush-vm-ip
Microsoft.Network/publicIpAddre...
OK
Operation details
Next steps
Setup auto-shutdown Recommended
Monitor VM health, performance and network dependencies Recommended
Run a script inside the virtual machine Recommended
Go to resource
Create another VM
Give feedback
Cost Management
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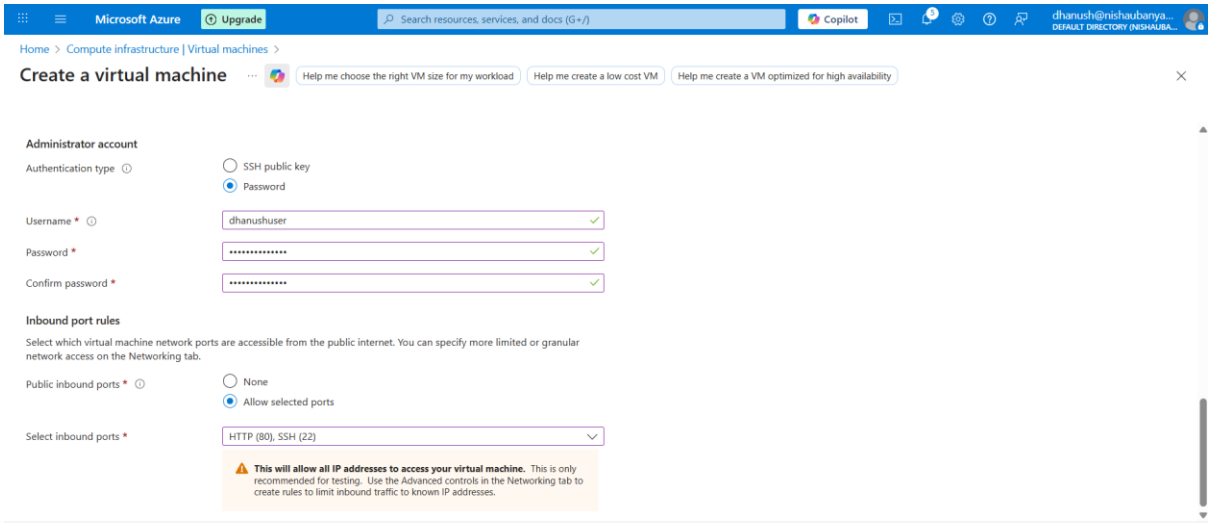
VM 2

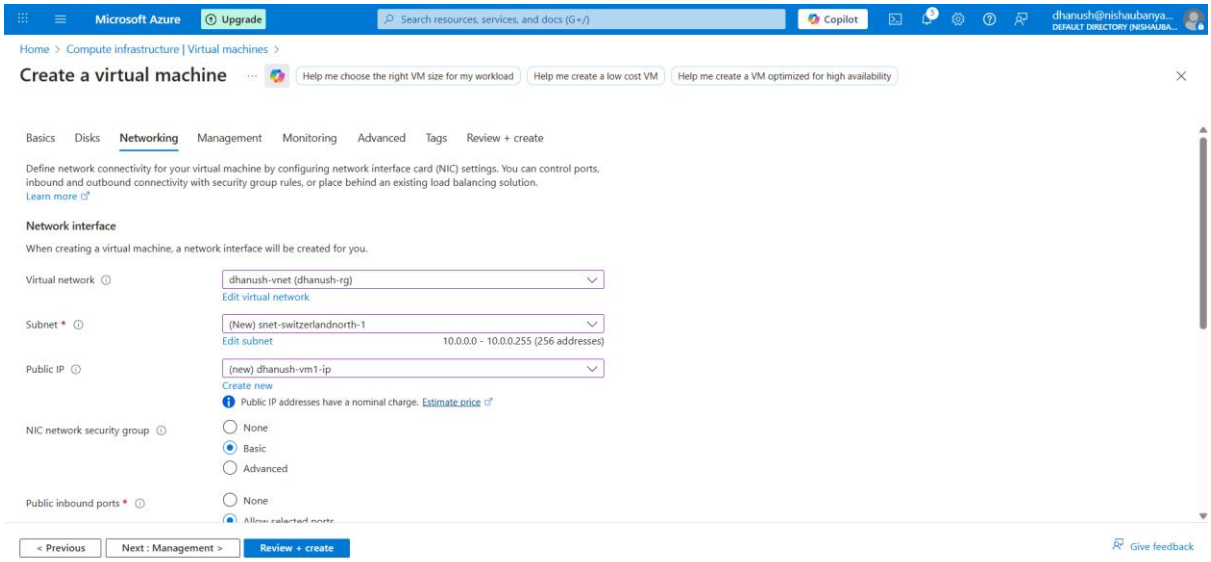

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DEFAULT DIRECTORY (NISHAUBA...
Home >
CreateVm-almalinux.almalinux-x86_64
Deployment
Search
Overview
Inputs
Outputs
Template
Delete
Cancel
Your deployme
Deployment name: Creat
Subscription: Azure subs
Resource group: dhanus
Deployment details
Resource
dhanush-vm
dhanush-vm707_z1
dhanush-vm-nsg
dhanush-vm-ip
Next steps
Setup auto-shutdown
Monitor VM health, perfor
Run a script inside the virt
Go to resource
Create another VM
Give feedback
Search history
Clear all
vm
resource
applic
vmss
loa
Recent services
Virtual machines
Resource groups
Application gateways
Virtual machine scal...
Load balancers
Virtual networks
Cost Management
Kubernetes services
Recent resources
dhanush-vm
Virtual machine
dhanush-rg
Resource group
Searching all subscriptions.
Give feedback
d70c5cd6b64f
peration details
peration details
peration details
peration details
peration details
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Set up cost alerts >
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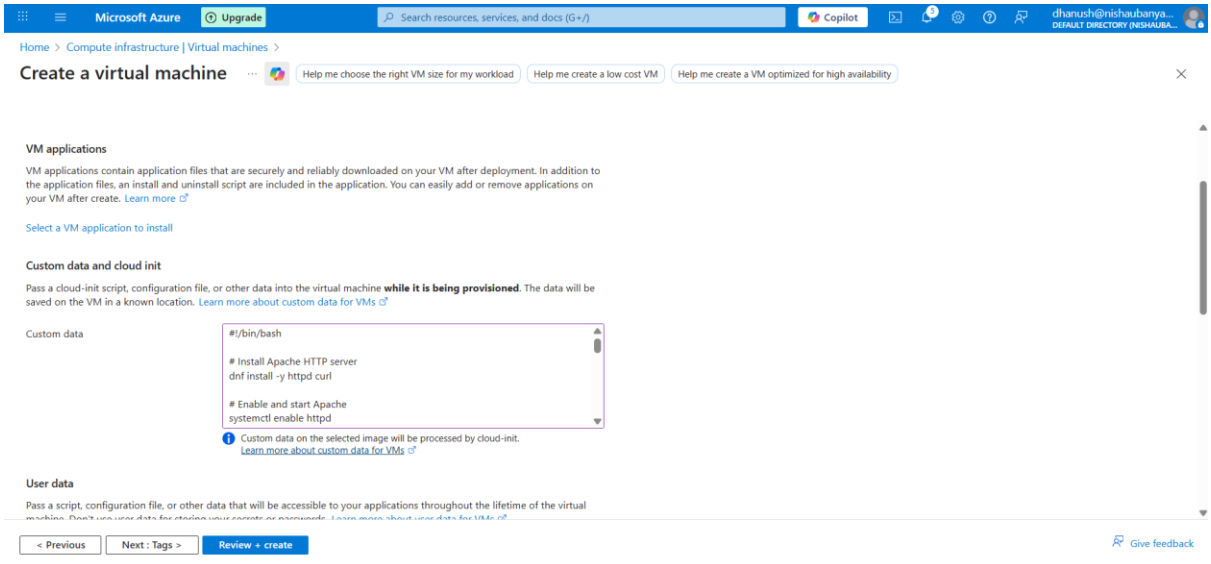

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DEFAULT DIRECTORY (NISHAUBA...
Home > Compute infrastructure | Virtual machines >
Create a virtual machine
Help me choose the right VM size for my workload
Help me create a low cost VM
Help me create a VM optimized for high availability
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription * Azure subscription 1
Resource group * dhanush-rg
Create new
Instance details
Virtual machine name * dhanush-vm1
Region * (Europe) Switzerland North
Deploy to an Azure Extended Zone
Availability options Availability zone
Zone options
Self-selected zone
Choose up to 3 availability zones, one VM per zone
Azure-selected zone (Preview)
Let Azure assign the best zone for your needs
Availability zone * Zone 2
You can now select multiple zones. Selecting multiple zones will create one VM per zone. Learn more
< Previous
Next : Disks >
Review + create
Give feedback

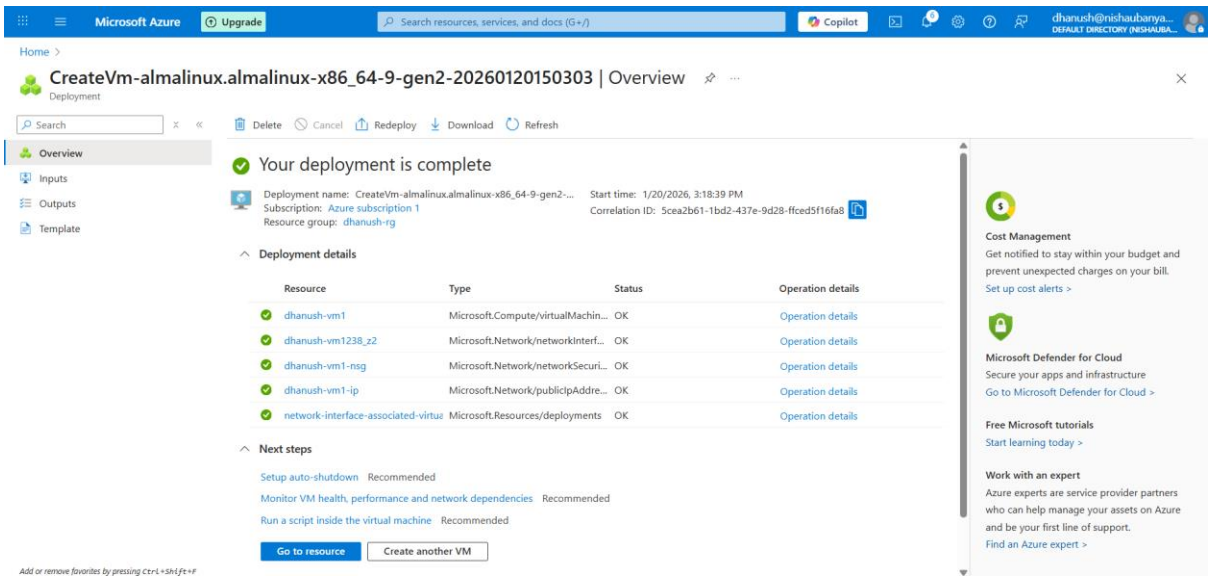

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DEFAULT DIRECTORY (NISHAUBA...
Home > Compute infrastructure | Virtual machines >
Create a virtual machine
Help me choose the right VM size for my workload
Help me create a low cost VM
Help me create a VM optimized for high availability
Subscription * Azure subscription 1
Resource group * dhanush-rg
Create new
Instance details
Virtual machine name * dhanush-vm1
Region * (Europe) Switzerland North
Deploy to an Azure Extended Zone
Availability options Availability zone
Zone options
Self-selected zone
Choose up to 3 availability zones, one VM per zone
Azure-selected zone (Preview)
Let Azure assign the best zone for your needs
Availability zone * Zone 2
You can now select multiple zones. Selecting multiple zones will create one VM per zone. Learn more
< Previous
Next : Disks >
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Give feedback

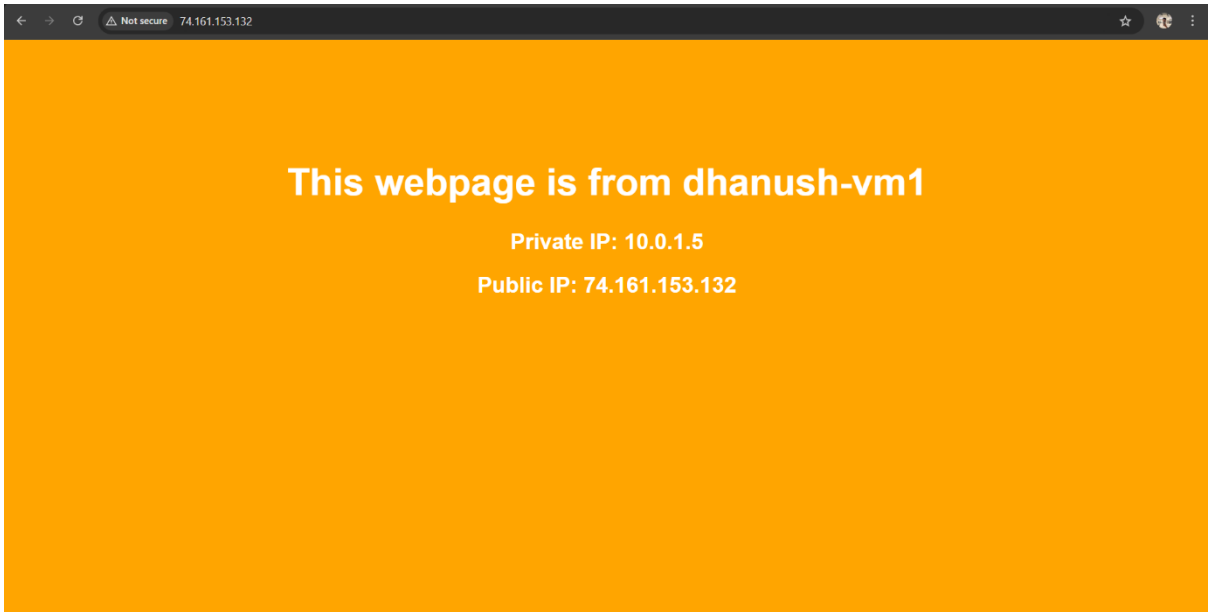

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DEFAULT DIRECTORY (NISHAUBA...
Home > Compute infrastructure | Virtual machines >
Create a virtual machine
Help me choose the right VM size for my workload
Help me create a low cost VM
Help me create a VM optimized for high availability
Security type
Trusted launch virtual machines
Configure security features
Image *
AlmaLinux OS 9 - x64 Gen2 (free services eligible)
See all images | Configure VM generation
VM architecture
Arm64
x64
Arm64 is not supported with the selected image.
Run with Azure Spot discount
You are in the free trial period. Costs associated with this VM can be covered by any remaining credits on your subscription.
Learn more
Size *
Standard_D2s_v3 - 2 vcpus, 8 GiB memory ($96.36)
See all sizes
Enable Hibernation
Hibernate does not currently support Trusted launch and Confidential virtual machines for Linux images. Learn more
< Previous
Next : Disks >
Review + create
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DEFAULT DIRECTORY (NISHAUBA...
Home > Compute infrastructure | Virtual machines >
Create a virtual machine
Help me choose the right VM size for my workload
Help me create a low cost VM
Help me create a VM optimized for high availability
Administrator account
Authentication type
SSH public key
Password
Username *
dhanushuser
Password *
Confirm password *
Inbound port rules
Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.
Public inbound ports *
None
Allow selected ports
Select inbound ports *
HTTP (80), SSH (22)
This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.
< Previous
Next : Disks >
Review + create
Give feedback


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DEFAULT DIRECTORY (NISHAUBA...
Home > Compute infrastructure | Virtual machines >
Create a virtual machine
Help me choose the right VM size for my workload
Help me create a low cost VM
Help me create a VM optimized for high availability
Basics
Disks
Networking
Management
Monitoring
Advanced
Tags
Review + create
Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.
Learn more
Network interface
When creating a virtual machine, a network interface will be created for you.
Virtual network
dhanush-vnet (dhanush-rg)
Edit virtual network
Subnet *
(New) snet-switzerlandnorth-1
Edit subnet
10.0.0.0 - 10.0.0.255 (256 addresses)
Public IP
(new) dhanush-vm1-ip
Create new
Public IP addresses have a nominal charge. Estimate price
NIC network security group
None
Basic
Advanced
Public inbound ports *
None
< Previous
Next : Management >
Review + create
Give feedback


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DEFAULT DIRECTORY (NISHAUBA...
Home > Compute infrastructure | Virtual machines >
Create a virtual machine
Help me choose the right VM size for my workload
Help me create a low cost VM
Help me create a VM optimized for high availability
VM applications
VM applications contain application files that are securely and reliably downloaded on your VM after deployment. In addition to the application files, an install and uninstall script are included in the application. You can easily add or remove applications on your VM after create. Learn more
Select a VM application to install
Custom data and cloud init
Pass a cloud-init script, configuration file, or other data into the virtual machine while it is being provisioned. The data will be saved on the VM in a known location. Learn more about custom data for VMs
Custom data
#!/bin/bash
Install Apache HTTP server
dnf install -y httpd curl
Enable and start Apache
systemctl enable httpd
Custom data on the selected image will be processed by cloud-init.
Learn more about custom data for VMs
User data
Pass a script, configuration file, or other data that will be accessible to your applications throughout the lifetime of the virtual
< Previous
Next : Tags >
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Copilot
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DEFAULT DIRECTORY (NISHAUBA...
Home >
CreateVm-almalinux.almalinux-x86_64-9-gen2-20260120150303 | Overview
Deployment
Search
Overview
Inputs
Outputs
Template
Delete
Cancel
Redeploy
Download
Refresh
Your deployment is complete
Deployment name: CreateVm-almalinux.almalinux-x86_64-9-gen2-...
Subscription: Azure subscription 1
Resource group: dhanush-rg
Start time: 1/20/2026, 3:18:39 PM
Correlation ID: 5cea2b61-1bd2-437e-9d28-ffced5f16fa8
Deployment details
Resource
Type
Status
Operation details
dhanush-vm1
Microsoft.Compute/virtualMachin...
OK
Operation details
dhanush-vm1238_z2
Microsoft.Network/networkInterf...
OK
Operation details
dhanush-vm1-nsg
Microsoft.Network/networkSecuri...
OK
Operation details
dhanush-vm1-ip
Microsoft.Network/publicIpAddre...
OK
Operation details
network-interface-associated-virtua
Microsoft.Resources/deployments
OK
Operation details
Next steps
Setup auto-shutdown Recommended
Monitor VM health, performance and network dependencies Recommended
Run a script inside the virtual machine Recommended
Go to resource
Create another VM
Add or remove favorites by pressing Ctrl+Shift+F
Cost Management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >
Microsoft Defender for Cloud
Secure your apps and infrastructure
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STEP 4 : Create Self-Signed PFX Certificate

Run in Cloud Shell or local Linux:

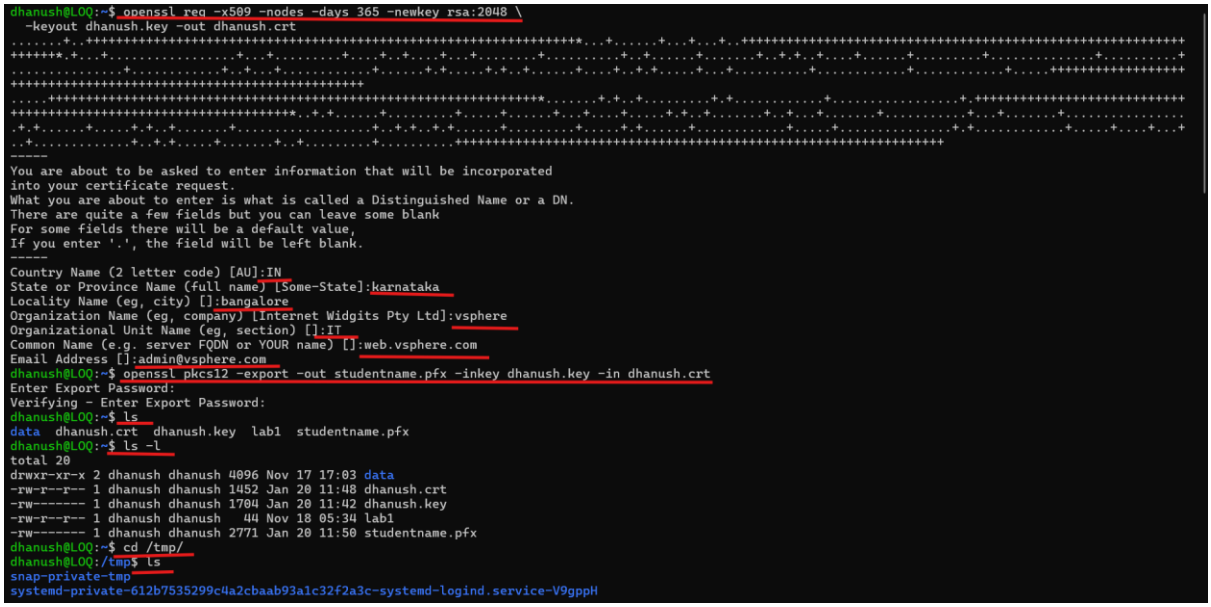
openssl req -x509 -nodes -days 365 -newkey rsa:2048 \

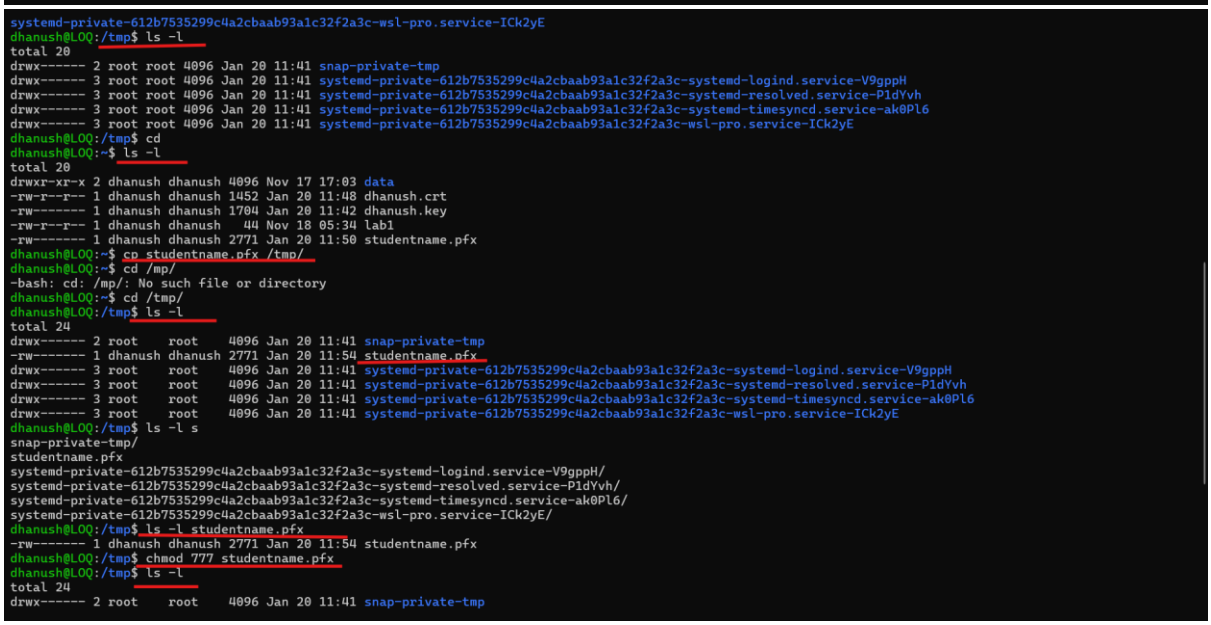
-keyout studentname.key -out studentname.crt

Export PFX:

openssl pkcs12 -export -out studentname.pfx -inkey studentname.key -in studentname.crt

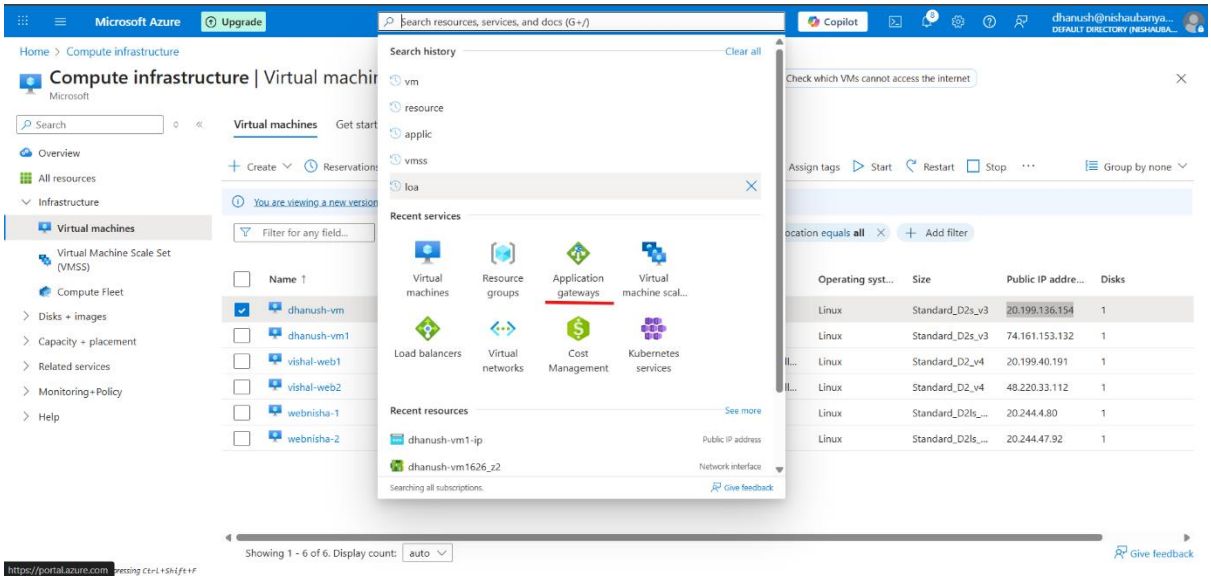
Enter an export password → note it.

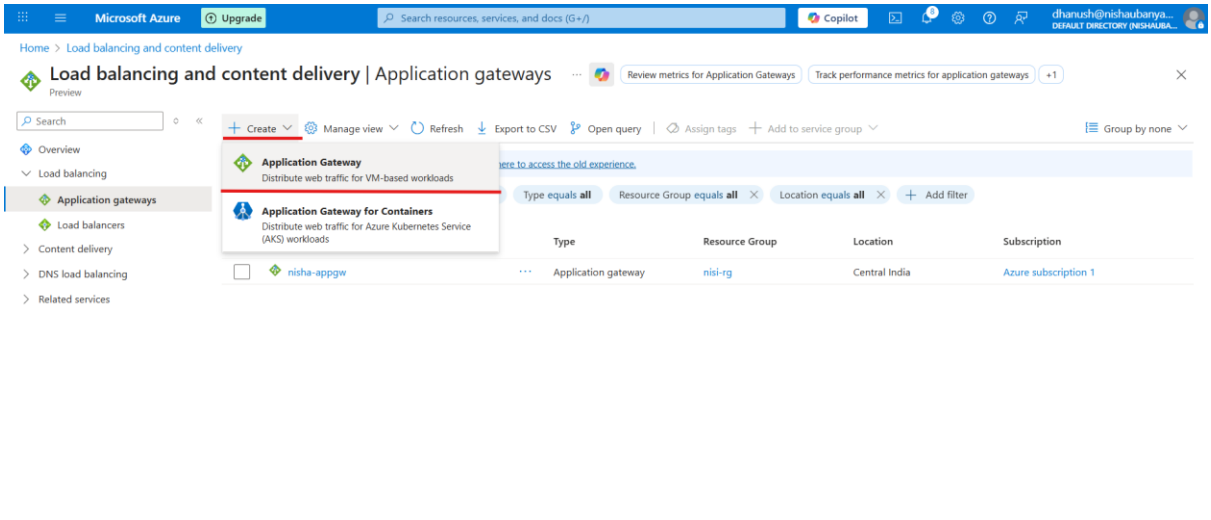


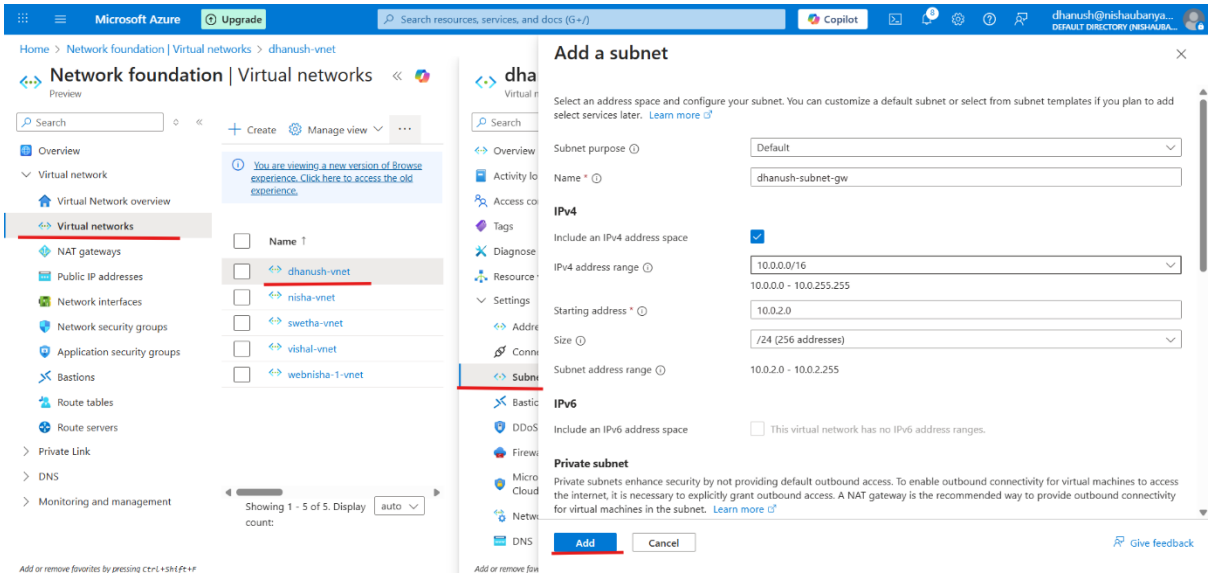


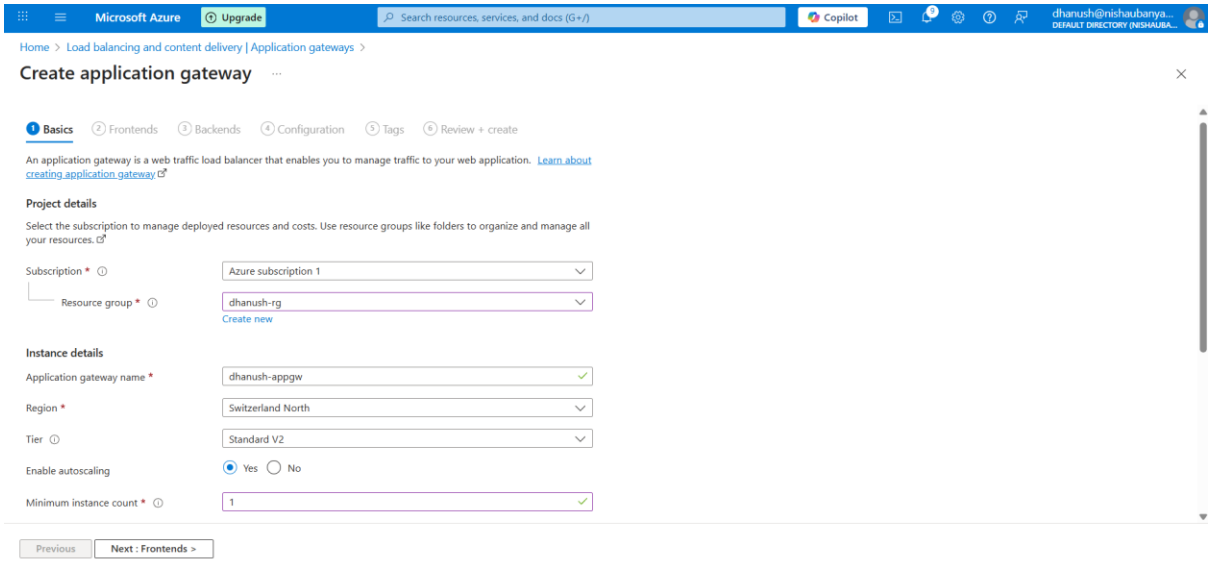


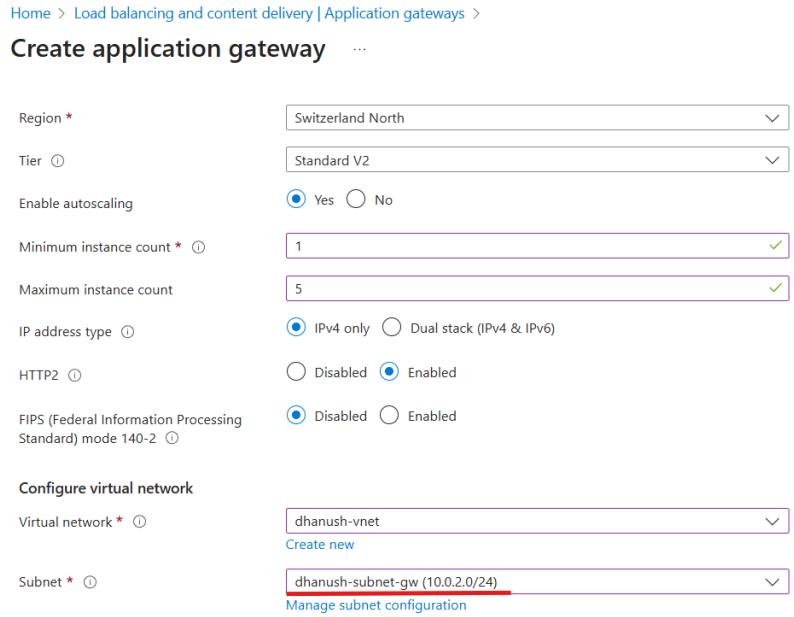
STEP 5 :Create Application Gateway

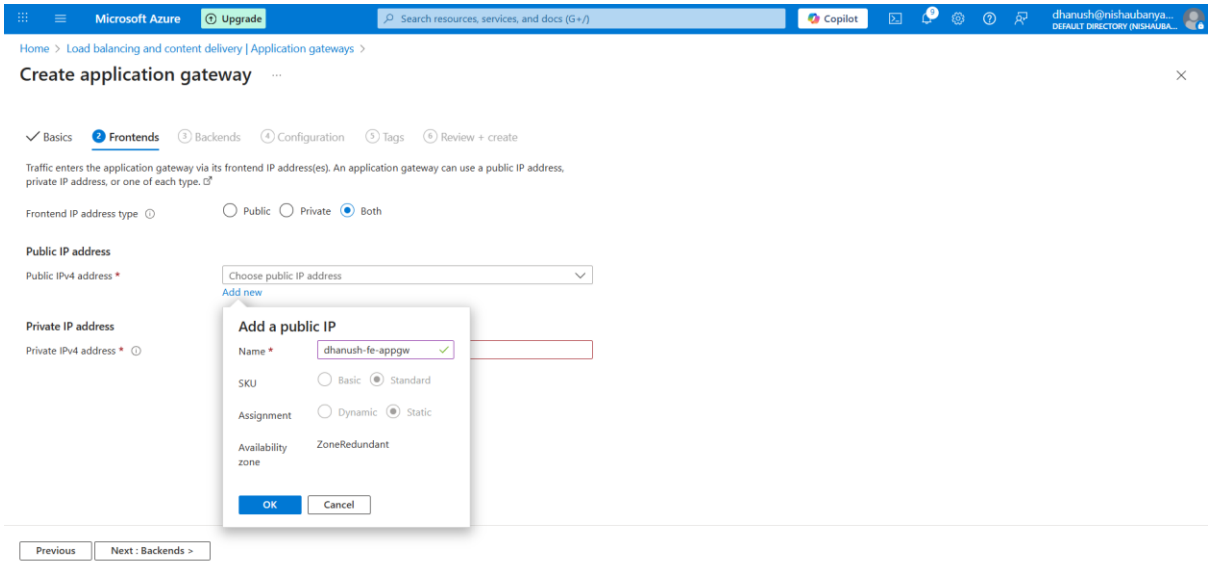

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Home > Compute infrastructure
Compute infrastructure | Virtual machi
Microsoft
Search history
Clear all
vm
resource
applic
vmss
loa
Recent services
Virtual machines
Resource groups
Application gateways
Virtual machine scal...
Load balancers
Virtual networks
Cost Management
Kubernetes services
Recent resources
See more
dhanush-vm1-ip
Public IP address
dhanush-vm1626_z2
Network interface
Searching all subscriptions.
Give feedback
Overview
All resources
Infrastructure
Virtual machines
Virtual Machine Scale Set (VMSS)
Compute Fleet
Disks + images
Capacity + placement
Related services
Monitoring+Policy
Help
Virtual machines
Get start
Create
Reservations
Assign tags
Start
Restart
Stop
Group by none
Add filter
Name
Operating syst...
Size
Public IP addre...
Disks
dhanush-vm
dhanush-vm1
vishal-web1
vishal-web2
webnisha-1
webnisha-2
Linux
Standard_D2s_v3
20.199.136.154
74.161.153.132
Standard_D2_v4
20.199.40.191
48.220.33.112
Standard_D2ls_...
20.244.4.80
20.244.47.92
Showing 1 - 6 of 6. Display count: auto
Give feedback

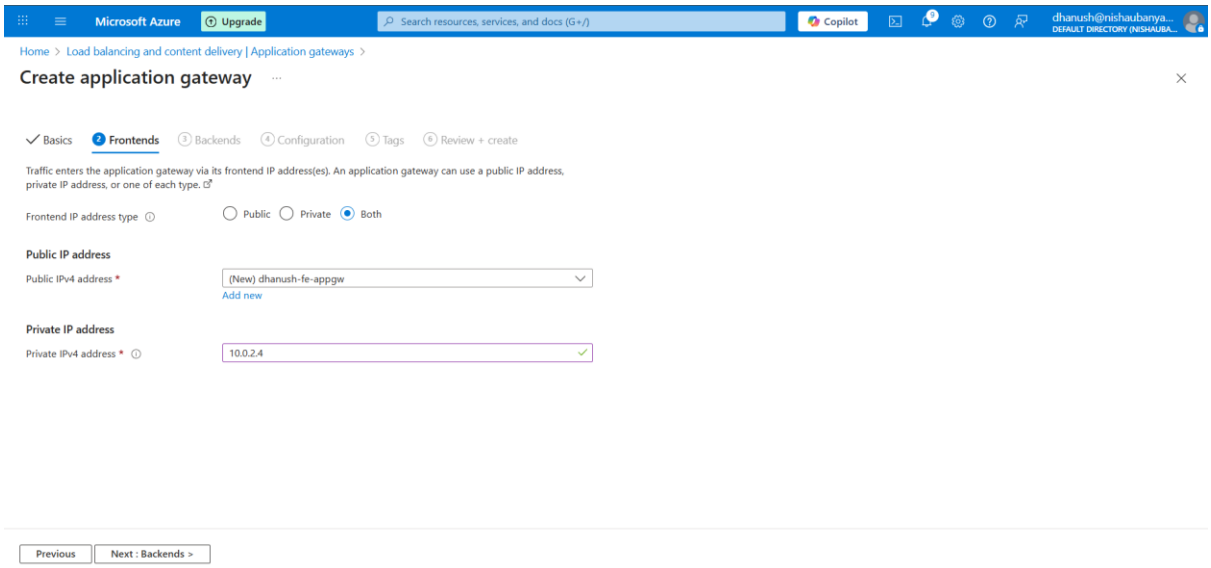

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Copilot
Home > Load balancing and content delivery
Load balancing and content delivery | Application gateways
Preview
Review metrics for Application Gateways
Track performance metrics for application gateways
+1
Create
Manage view
Refresh
Export to CSV
Open query
Assign tags
Add to service group
Group by none
Application Gateway
Distribute web traffic for VM-based workloads
Application Gateway for Containers
Distribute web traffic for Azure Kubernetes Service (AKS) workloads
Type equals all
Resource Group equals all
Location equals all
Add filter
Overview
Load balancing
Application gateways
Load balancers
Content delivery
DNS load balancing
Related services
Type
Resource Group
Location
Subscription
nisha-appgw
Application gateway
nisi-rg
Central India
Azure subscription 1
Showing 1 - 1 of 1. Display count: auto
Give feedback

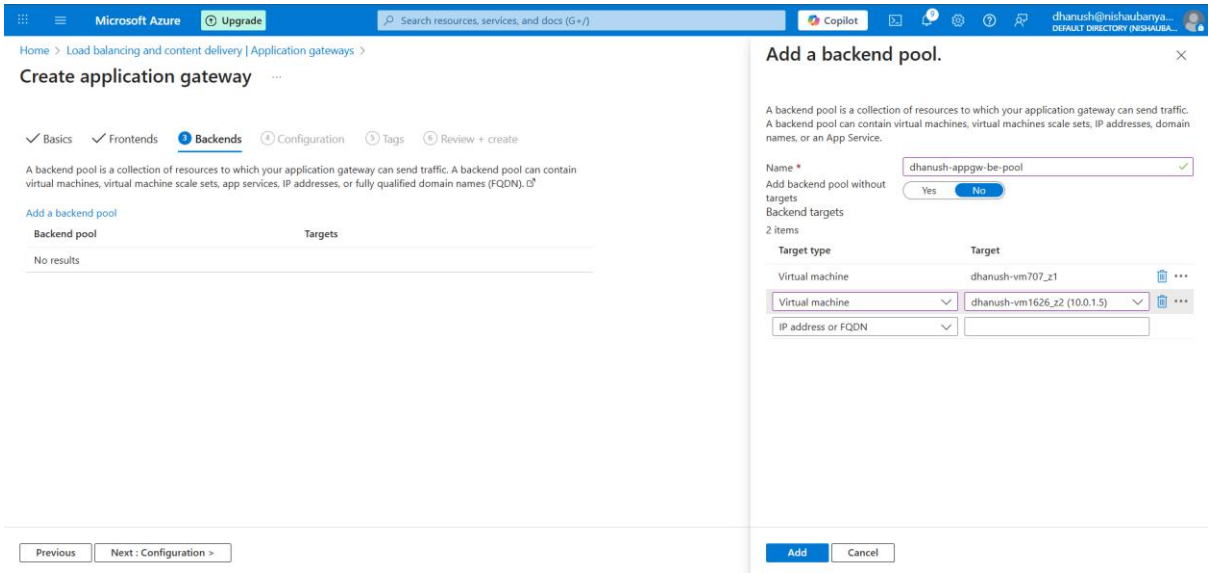

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Copilot
Home > Network foundation | Virtual networks > dhanush-vnet
Network foundation | Virtual networks
Preview
Create
Manage view
You are viewing a new version of Browse experience. Click here to access the old experience.
Overview
Virtual network
Virtual Network overview
Virtual networks
NAT gateways
Public IP addresses
Network interfaces
Network security groups
Application security groups
Bastions
Route tables
Route servers
Private Link
DNS
Monitoring and management
Name
dhanush-vnet
nisha-vnet
swetha-vnet
vishal-vnet
webnisha-1-vnet
Showing 1 - 5 of 5. Display count: auto
Add a subnet
Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. Learn more
Subnet purpose
Default
Name *
dhanush-subnet-gw
IPv4
Include an IPv4 address space
IPv4 address range
10.0.0.0/16
10.0.0.0 - 10.0.255.255
Starting address *
10.0.2.0
Size
/24 (256 addresses)
Subnet address range
10.0.2.0 - 10.0.2.255
IPv6
Include an IPv6 address space
This virtual network has no IPv6 address ranges.
Private subnet
Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound connectivity for virtual machines in the subnet. Learn more
Add
Cancel
Give feedback


Microsoft Azure
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Search resources, services, and docs (G+/)
Copilot
dhanush@nishaubanya...
DEFAULT DIRECTORY (NISHAUBA...
Home > Load balancing and content delivery | Application gateways >
Create application gateway
Basics
Frontends
Backends
Configuration
Tags
Review + create
An application gateway is a web traffic load balancer that enables you to manage traffic to your web application. Learn about creating application gateway
Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Azure subscription 1
Resource group *
dhanush-rg
Create new
Instance details
Application gateway name *
dhanush-appgw
Region *
Switzerland North
Tier
Standard V2
Enable autoscaling
Yes
No
Minimum instance count *
1
Previous
Next : Frontends >

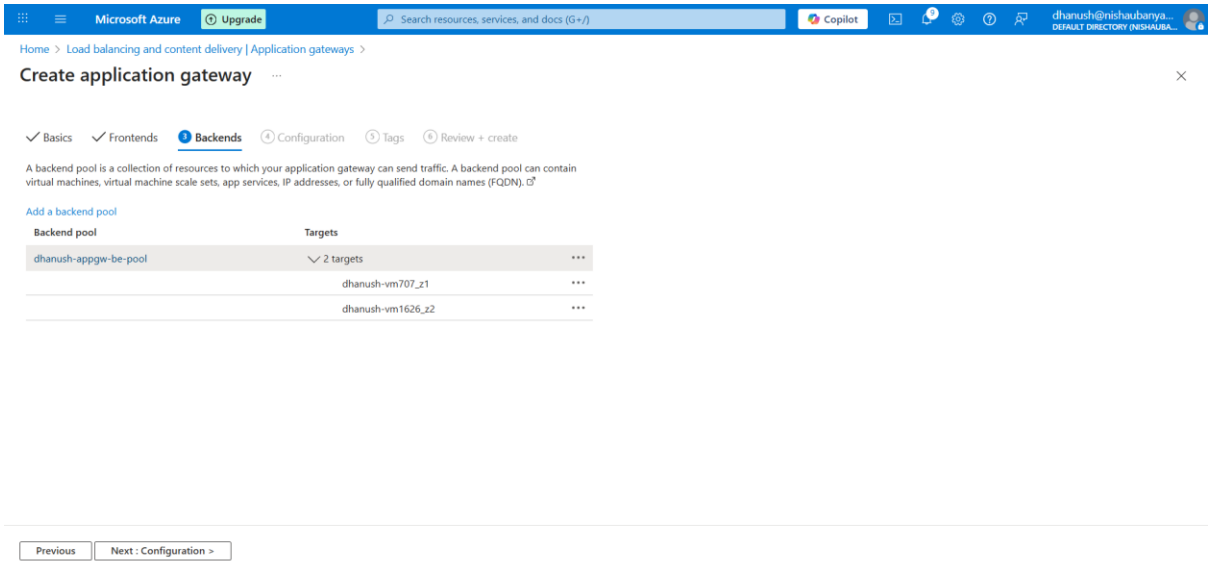

Home > Load balancing and content delivery | Application gateways >
Create application gateway
Region *
Switzerland North
Tier
Standard V2
Enable autoscaling
Yes
No
Minimum instance count *
1
Maximum instance count
5
IP address type
IPv4 only
Dual stack (IPv4 & IPv6)
HTTP2
Disabled
Enabled
FIPS (Federal Information Processing Standard) mode 140-2
Disabled
Enabled
Configure virtual network
Virtual network *
dhanush-vnet
Create new
Subnet *
dhanush-subnet-gw (10.0.2.0/24)
Manage subnet configuration
Previous
Next : Frontends >


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Create application gateway
Basics
Frontends
Backends
Configuration
Tags
Review + create
Traffic enters the application gateway via its frontend IP address(es). An application gateway can use a public IP address, private IP address, or one of each type.
Frontend IP address type
Public
Private
Both
Public IP address
Public IPv4 address *
Choose public IP address
Add new
Private IP address
Private IPv4 address *
Add a public IP
Name *
dhanush-fe-appgw
SKU
Basic
Standard
Assignment
Dynamic
Static
Availability zone
ZoneRedundant
OK
Cancel
Previous
Next : Backends >

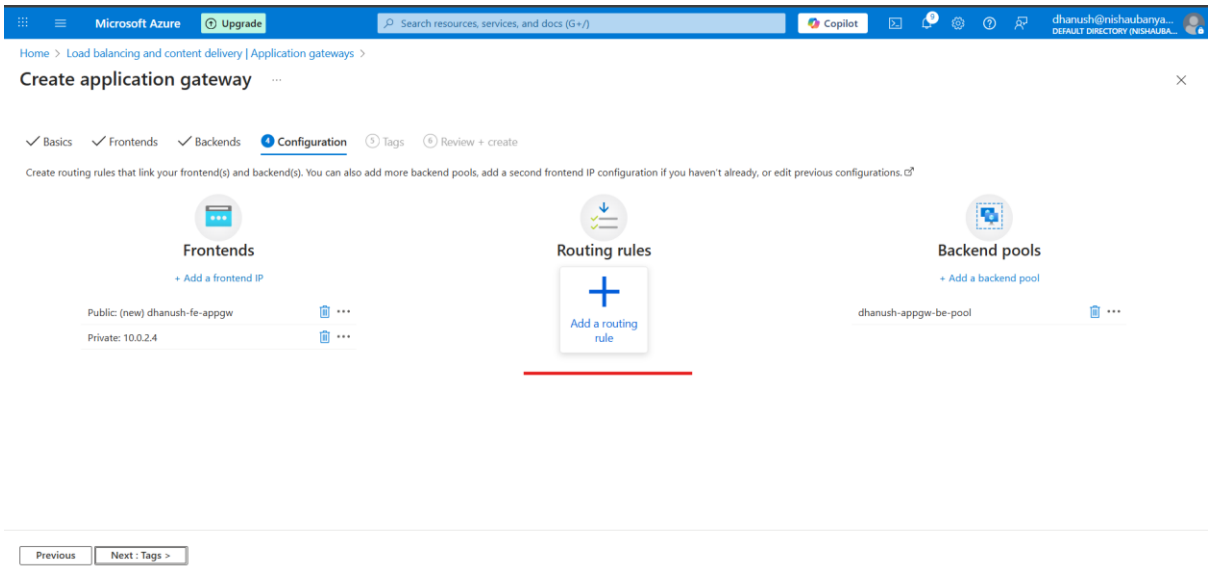

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DEFAULT DIRECTORY (NISHAUBA...
Home > Load balancing and content delivery | Application gateways >
Create application gateway
Basics
Frontends
Backends
Configuration
Tags
Review + create
Traffic enters the application gateway via its frontend IP address(es). An application gateway can use a public IP address, private IP address, or one of each type.
Frontend IP address type
Public
Private
Both
Public IP address
Public IPv4 address *
(New) dhanush-fe-appgw
Add new
Private IP address
Private IPv4 address *
10.0.2.4
Previous
Next : Backends >


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DEFAULT DIRECTORY (NISHAUBA...
Home > Load balancing and content delivery | Application gateways >
Create application gateway
Basics
Frontends
Backends
Configuration
Tags
Review + create
A backend pool is a collection of resources to which your application gateway can send traffic. A backend pool can contain virtual machines, virtual machine scale sets, app services, IP addresses, or fully qualified domain names (FQDN).
Add a backend pool
Backend pool
Targets
No results
Previous
Next : Configuration >

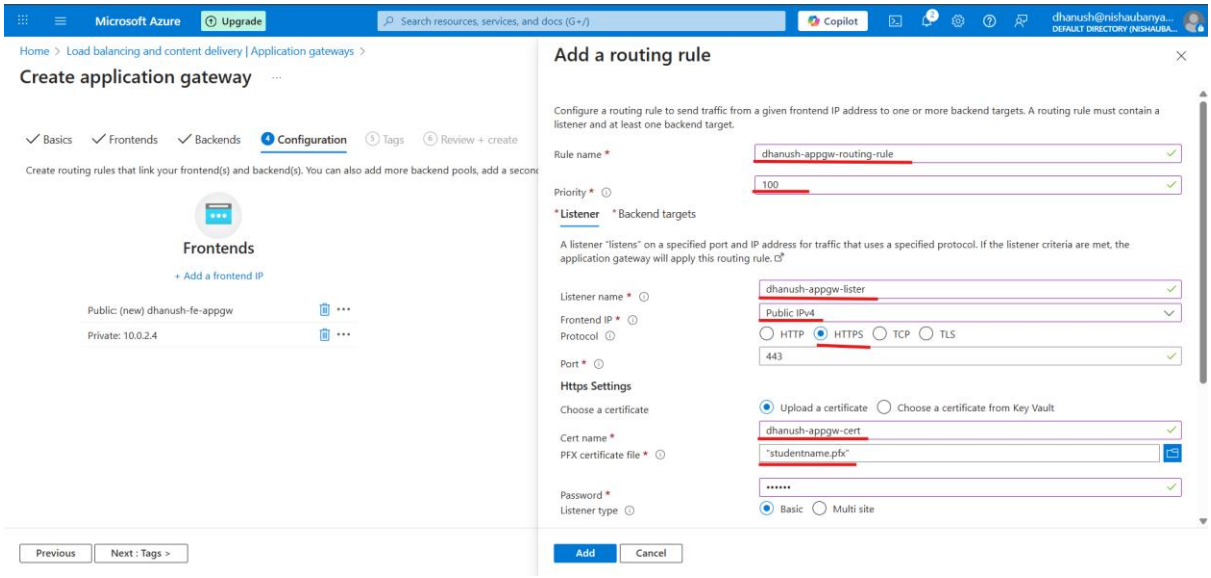
Add a backend pool.
A backend pool is a collection of resources to which your application gateway can send traffic. A backend pool can contain virtual machines, virtual machines scale sets, IP addresses, domain names, or an App Service.
Name *
dhanush-appgw-be-pool
Add backend pool without targets
Yes
No
Backend targets
2 items
Target type
Target
Virtual machine
dhanush-vm707_z1
Virtual machine
dhanush-vm1626_z2 (10.0.1.5)
IP address or FQDN
Add
Cancel

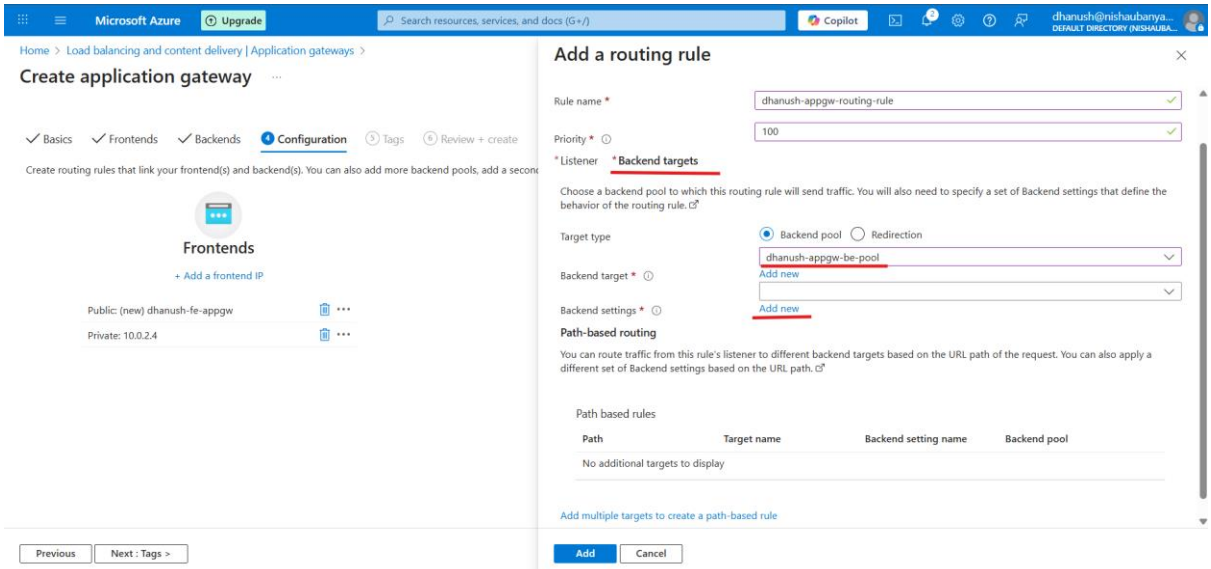


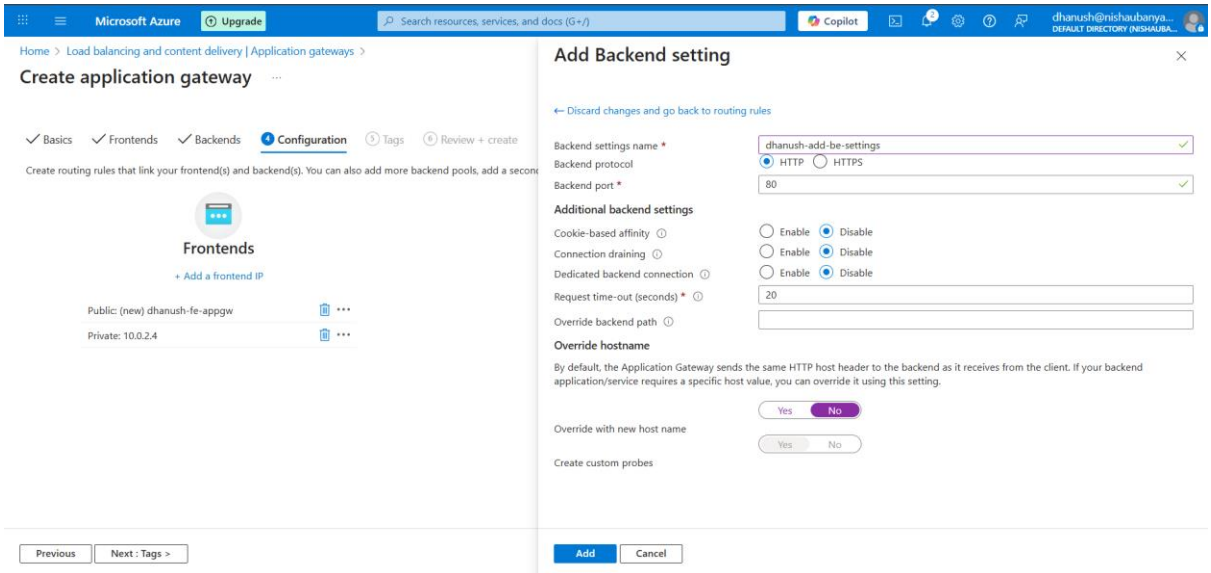
Previous | Next : Configuration >

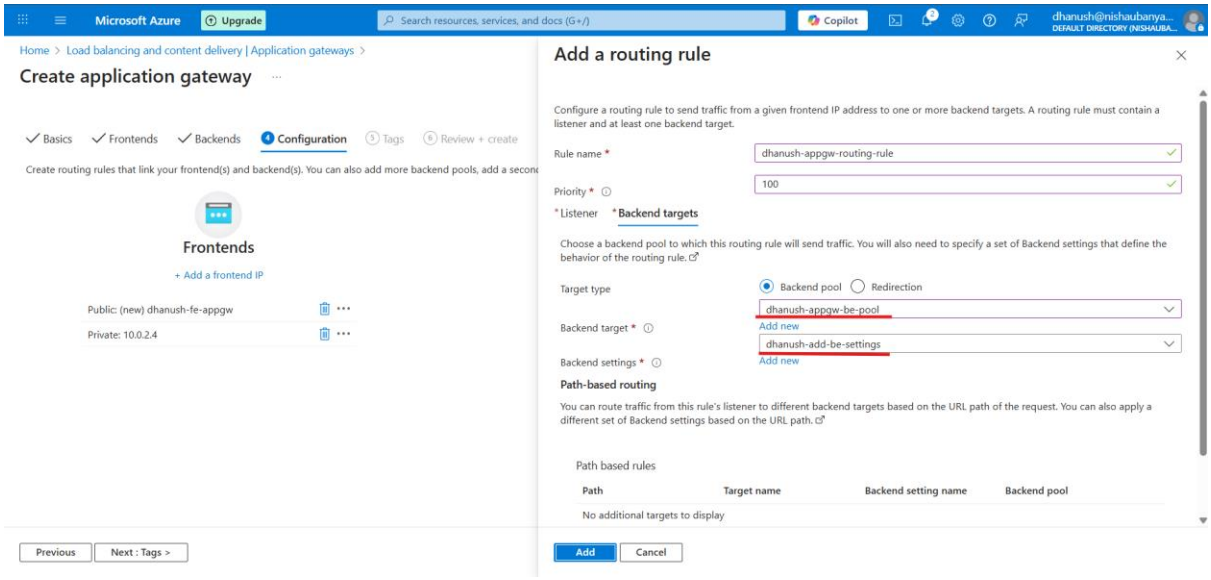


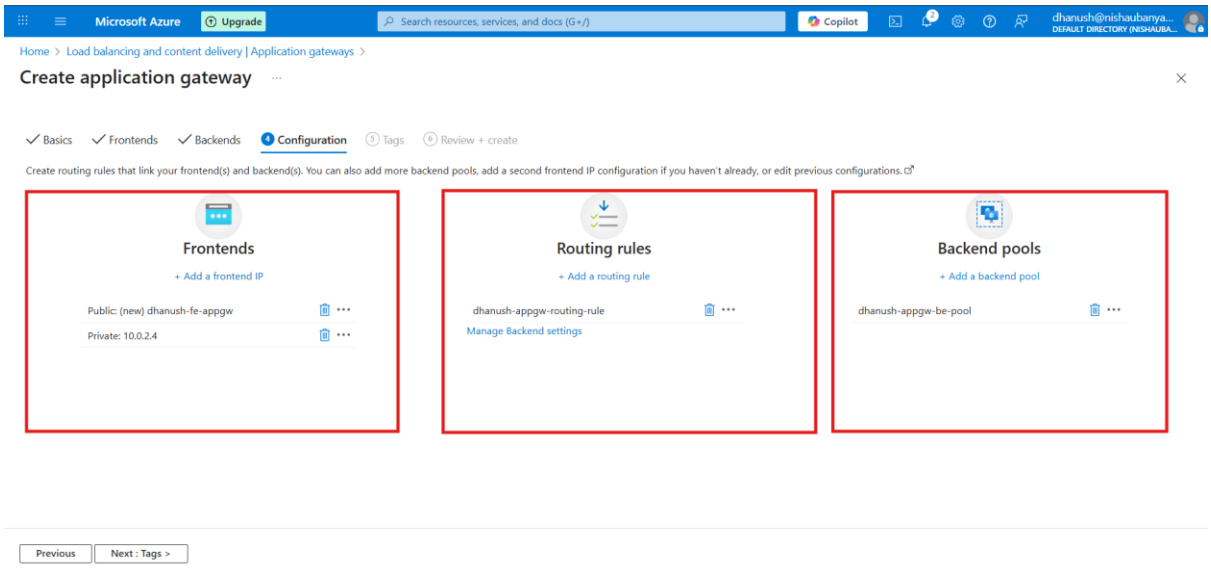
Previous | Next : Tags >

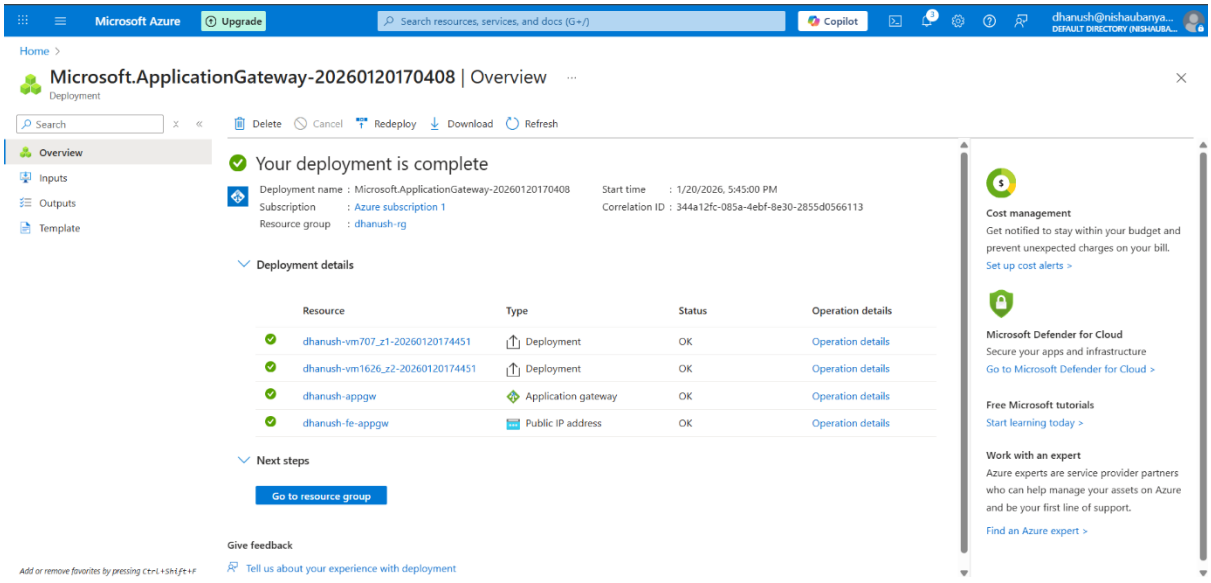

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DEFAULT DIRECTORY (NISHAUBA...
Home > Load balancing and content delivery | Application gateways >
Create application gateway
Basics
Frontends
Backends
Configuration
Tags
Review + create
Create routing rules that link your frontend(s) and backend(s). You can also add more backend pools, add a secon
Frontends
+ Add a frontend IP
Public: (new) dhanush-fe-appgw
Private: 10.0.2.4
Previous
Next : Tags >
Add a routing rule
Configure a routing rule to send traffic from a given frontend IP address to one or more backend targets. A routing rule must contain a listener and at least one backend target.
Rule name *
dhanush-appgw-routing-rule
Priority *
100
*Listener
*Backend targets
A listener "listens" on a specified port and IP address for traffic that uses a specified protocol. If the listener criteria are met, the application gateway will apply this routing rule.
Listener name *
dhanush-appgw-lister
Frontend IP *
Public IPv4
Protocol
HTTP
HTTPS
TCP
TLS
Port *
443
Https Settings
Choose a certificate
Upload a certificate
Choose a certificate from Key Vault
Cert name *
dhanush-appgw-cert
PFX certificate file *
"studentname.pfx"
Password *
Listener type
Basic
Multi site
Add
Cancel


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Home > Load balancing and content delivery | Application gateways >
Create application gateway
Basics
Frontends
Backends
Configuration
Tags
Review + create
Create routing rules that link your frontend(s) and backend(s). You can also add more backend pools, add a secon
Frontends
+ Add a frontend IP
Public: (new) dhanush-fe-appgw
Private: 10.0.2.4
Previous
Next : Tags >
Add a routing rule
Rule name *
dhanush-appgw-routing-rule
Priority *
100
*Listener
*Backend targets
Choose a backend pool to which this routing rule will send traffic. You will also need to specify a set of Backend settings that define the behavior of the routing rule.
Target type
Backend pool
Redirection
Backend target *
dhanush-appgw-be-pool
Add new
Backend settings *
Add new
Path-based routing
You can route traffic from this rule's listener to different backend targets based on the URL path of the request. You can also apply a different set of Backend settings based on the URL path.
Path based rules
Path
Target name
Backend setting name
Backend pool
No additional targets to display
Add multiple targets to create a path-based rule
Add
Cancel

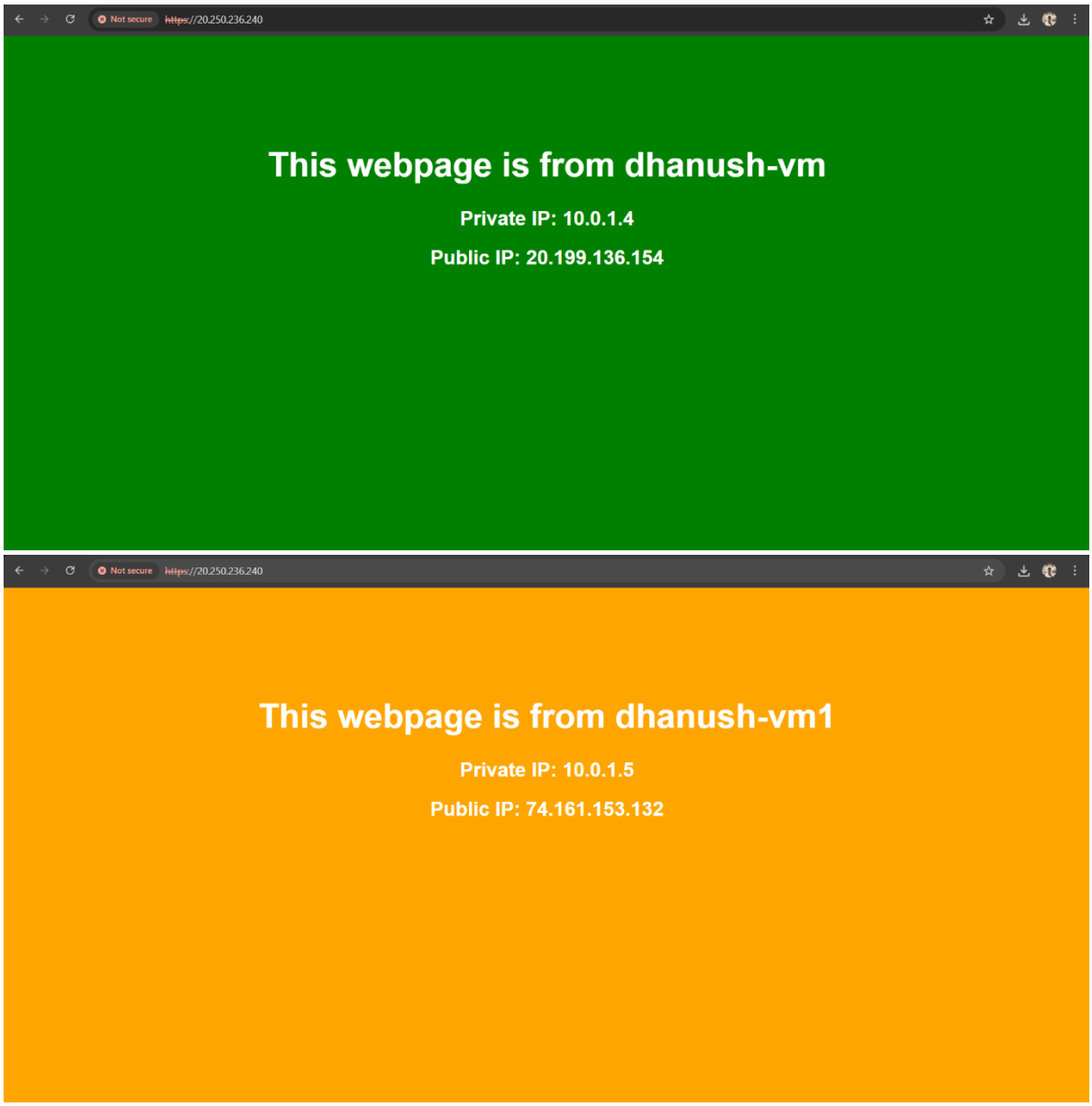

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Home > Load balancing and content delivery | Application gateways >
Create application gateway
Basics
Frontends
Backends
Configuration
Tags
Review + create
Create routing rules that link your frontend(s) and backend(s). You can also add more backend pools, add a secon
Frontends
+ Add a frontend IP
Public: (new) dhanush-fe-appgw
Private: 10.0.2.4
Previous
Next : Tags >
Add Backend setting
← Discard changes and go back to routing rules
Backend settings name *
dhanush-add-be-settings
Backend protocol
HTTP
HTTPS
Backend port *
80
Additional backend settings
Cookie-based affinity
Enable
Disable
Connection draining
Enable
Disable
Dedicated backend connection
Enable
Disable
Request time-out (seconds) *
20
Override backend path
Override hostname
By default, the Application Gateway sends the same HTTP host header to the backend as it receives from the client. If your backend application/service requires a specific host value, you can override it using this setting.
Override with new host name
Yes
No
Create custom probes
Add
Cancel


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DEFAULT DIRECTORY (NISHAUBA...
Home > Load balancing and content delivery | Application gateways >
Create application gateway
Basics
Frontends
Backends
Configuration
Tags
Review + create
Create routing rules that link your frontend(s) and backend(s). You can also add more backend pools, add a secon
Frontends
+ Add a frontend IP
Public: (new) dhanush-fe-appgw
Private: 10.0.2.4
Previous
Next : Tags >
Add a routing rule
Configure a routing rule to send traffic from a given frontend IP address to one or more backend targets. A routing rule must contain a listener and at least one backend target.
Rule name *
dhanush-appgw-routing-rule
Priority *
100
*Listener
*Backend targets
Choose a backend pool to which this routing rule will send traffic. You will also need to specify a set of Backend settings that define the behavior of the routing rule.
Target type
Backend pool
Redirection
Backend target *
dhanush-appgw-be-pool
Add new
Backend settings *
dhanush-add-be-settings
Add new
Path-based routing
You can route traffic from this rule's listener to different backend targets based on the URL path of the request. You can also apply a different set of Backend settings based on the URL path.
Path based rules
Path
Target name
Backend setting name
Backend pool
No additional targets to display
Add
Cancel


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Copilot
dhanush@nishaubanya...
DEFAULT DIRECTORY (NISHAUBA...
Home > Load balancing and content delivery | Application gateways >
Create application gateway
Basics
Frontends
Backends
Configuration
Tags
Review + create
Create routing rules that link your frontend(s) and backend(s). You can also add more backend pools, add a second frontend IP configuration if you haven't already, or edit previous configurations.
Frontends
+ Add a frontend IP
Public: (new) dhanush-fe-appgw
Private: 10.0.2.4
Routing rules
+ Add a routing rule
dhanush-appgw-routing-rule
Manage Backend settings
Backend pools
+ Add a backend pool
dhanush-appgw-be-pool
Previous
Next : Tags >


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
dhanush@nishaubanya...
DEFAULT DIRECTORY (NISHAUBA...
Home >
Microsoft.ApplicationGateway-20260120170408 | Overview
Deployment
Search
Overview
Inputs
Outputs
Template
Delete
Cancel
Redeploy
Download
Refresh
Your deployment is complete
Deployment name : Microsoft.ApplicationGateway-20260120170408
Subscription : Azure subscription 1
Resource group : dhanush-rg
Start time : 1/20/2026, 5:45:00 PM
Correlation ID : 344a12fc-085a-4ebf-8e30-2855d0566113
Deployment details
Resource
Type
Status
Operation details
dhanush-vm707_z1-20260120174451
Deployment
OK
Operation details
dhanush-vm1626_z2-20260120174451
Deployment
OK
Operation details
dhanush-appgw
Application gateway
OK
Operation details
dhanush-fe-appgw
Public IP address
OK
Operation details
Next steps
Go to resource group
Give feedback
Tell us about your experience with deployment
Add or remove favorites by pressing Ctrl+Shift+F
Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >
Microsoft Defender for Cloud
Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >
Free Microsoft tutorials
Start learning today >
Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
Find an Azure expert >

OUTPUT :



---

1. Create Resource Group

- **Name:** studentname-rg
- **Region:** East US (or nearest region)

---

2. Create Two Linux VMs (Web Servers)

1. Go to **Virtual Machines → Create**
    - **Name:** studentname-vm1
    - Image: Ubuntu LTS
    - Size: Standard_B1s
    - Authentication: Password/SSH
    - Networking: Place in **studentname-vnet** (create if not existing), subnet web-subnet
    - Allow inbound ports: HTTP (80), SSH (22)

Repeat for **studentname-vm2**.

2. **Install Apache & HTML content** on each VM:

SSH into studentname-vm1:

```
sudo apt update
sudo apt install -y apache2
echo "<h1>Welcome to Web Server 1</h1>" | sudo tee /var/www/html/index.html
```

SSH into studentname-vm2:

sudo apt update

sudo apt install -y apache2

echo "<h1>Welcome to Web Server 2</h1>" | sudo tee /var/www/html/index.html

✅ Test in browser with public IPs:

- http://<vm1-public-ip> → Web Server 1
- http://<vm2-public-ip> → Web Server 2

---

3. Create Virtual Network

- **Name:** studentname-vnet
- Subnets:
  - web-subnet → VMs
  - appgw-subnet → Application Gateway

---

4. Create Application Gateway

1. Go to **Create Resource → Application Gateway**
   - **Name:** studentname-appgw
   - **Tier:** Standard_v2

- **Frontend IP:** Public
- **VNet:** studentname-vnet
- **Subnet:** appgw-subnet

2. Configure **Backend Pool**:
   - Add targets = private IPs of studentname-vm1 and studentname-vm2
3. Configure **Backend Settings**:
   - Port: 80
   - Protocol: HTTP
4. Configure **Listener (HTTP)**:
   - Listener name: http-listener
   - Protocol: HTTP
   - Port: 80
5. Configure **Rule**:
   - Route all HTTP requests to backend pool.

✅ Test: http://<appgw-public-ip> → Should load balance between VM1 and VM2.

---

5. Create Self-Signed PFX Certificate

Run in Cloud Shell or local Linux:

```
openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
  -keyout studentname.key -out studentname.crt
```

Export PFX:

```
openssl pkcs12 -export -out studentname.pfx -inkey studentname.key -in studentname.crt
```

Enter an export password → note it.

---

6. Create HTTPS Listener

1. In **Application Gateway → Listeners → Add**
    - Name: https-listener
    - Protocol: HTTPS
    - Port: 443
    - Upload certificate: studentname.pfx + password
2. Add new **Routing Rule**:
    - Listener = https-listener
    - Backend Pool = studentname-vm1, studentname-vm2

- Backend Settings = Port 80

✅ Test:

- https://<appgw-public-ip> → Should load balance between VM1 and VM2

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7. Create DNS Zone

1. Go to **Create Resource → DNS Zone**
   - Name: studentname.com
   - Resource Group: studentname-rg
2. Inside DNS Zone, add **A record**:
   - Name: www
   - Type: A
   - Value: <appgw-public-ip>

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8. Test with DNS

- Browse: http://www.studentname.com → VM1 or VM2 (round-robin)
- Browse: https://www.studentname.com → Secure connection using uploaded certificate

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✅ Deliverables

1. Screenshots of:
    - VM HTML pages in browser
    - App Gateway backend pool with VMs
    - HTTP + HTTPS listeners
    - DNS Zone with A record
    - Successful http://www.studentname.com and https://www.studentname.com

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- VMs in web subnet
- App Gateway in appgw-subnet
- DNS Zone → App Gateway Public IP → Backend pool (VM1 & VM2)?